Subject

Object Oriented programming and Computer Graphics

Unit-V

2D, 3D Transformations and Projections

Contents:

- **2-D Transformations:** 2-D transformations - Translation, Scaling, Rotation and Shear, Rotation about an arbitrary point.
- **3-D Transformations:** 3-D transformations - Translation, Scaling, Rotation and Shear, Rotation about an arbitrary axis.
- **Projections:** Parallel (Oblique: Cavalier, Cabinet and orthographic: isometric, diametric, trimetric) and Perspective (Vanishing Points – 1 point, 2 point and 3 point)
- **Case Studies -**

  1) Affine Transformations Vlab (Vlab link: https://cse19-iiith.vlabs.ac.in/exp/affine-transformation/theory.html)

  2) Image augmentation in Deep learning

Introduction of Transformations

- Computer Graphics provide the facility of viewing object from different angles.
- The purpose of using computers for drawing is to provide facility to user to view the object from different angles, enlarging or reducing the scale or shape of object called as Transformation.
- We can have various types of transformations such as translation, scaling up or down, rotation, shearing, etc.
- When a transformation takes place on a 2D plane, it is called 2D transformation.

- There are two complementary points of view for describing object transformation.

1) Geometric Transformation:

- The object itself is transformed relative to the coordinate system or background. The mathematical statement of this viewpoint is defined by geometric transformations applied to each point of the object.

2) Coordinate Transformation:

- The object is held stationary while the coordinate system is transformed relative to the object. This effect is attained through the application of coordinate transformations.

- An example that helps to distinguish these two viewpoints:
- **The movement of an automobile against a scenic background we can simulate this by:**

1. Moving the automobile while keeping the background fixed-(**Geometric Transformation**)
2. We can keep the car fixed while moving the background scenery- (**Coordinate Transformation)**

Types of Transformations:

1. Translation
2. Scaling
3. Rotating
4. Reflection
5. Shearing

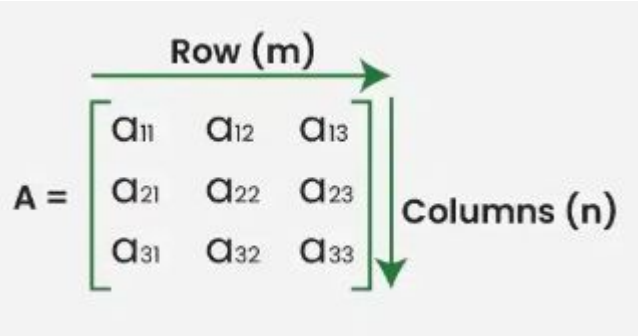
Basic 2D Transformation

- Translation
- Scaling
- Rotation

Matrices

- Matrix is a rectangular array of numbers, symbols, points, or characters each belonging to a specific row and column.
- A matrix is identified by its order which is given in the form of rows × and columns.
- The numbers, symbols, points, or characters present inside a matrix are called the elements of a matrix.
- The location of each element is given by the row and column it belongs to.
- A matrix is represented as

  [P]m×n



where P is the matrix,

m is the number of rows and n is the number of columns.

Order of Matrix

- Order of a Matrix tells about the number of rows and columns present in a matrix.

Example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times 2}, \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 6 \\ 4 & -2 & 5 \end{bmatrix}_{3\times 3}$

- **Operation on Matrices**

1. Addition,
2. Subtraction,
3. Scalar multiplication,
4. Multiplication.

Addition of Matrices

Matrix Addition of 2*2 Matrices

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

Example: Find the sum of $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ and $\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$

Solution:

Here, we have $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$

$$A + B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$\Rightarrow A + B = \begin{bmatrix} 1+2 & 2+3 \\ 4+6 & 5+7 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 10 & 12 \end{bmatrix}$$

2) Subtraction of Matrices

Example: Subtract $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ from $\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$.

Let us assume $A = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

$$A - B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$$

$$\Rightarrow A - B = \begin{bmatrix} 2-1 & 3-2 \\ 6-4 & 7-5 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Scalar Multiplication of Matrices

Example: Multiply 3 with $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$.

$$3[A] = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 4 & 3 \times 5 \end{bmatrix}$$

$$\Rightarrow 3[A] = \begin{bmatrix} 3 & 6 \\ 12 & 15 \end{bmatrix}$$

Multiplication of Matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} i & j \\ k & l \end{bmatrix} = \begin{bmatrix} a.i + b.k & a.j + b.l \\ c.i + d.k & c.j + d.l \end{bmatrix}$$

Example: Find the product of $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$ and $\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$

Solution:

$$\text{Let } A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 1 \times 2 + 2 \times 6 & 1 \times 3 + 2 \times 7 \\ 4 \times 2 + 5 \times 6 & 4 \times 3 + 5 \times 7 \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 14 & 17 \\ 38 & 47 \end{bmatrix}$$

Properties of Matrix Addition and Multiplication

Properties followed by Multiplication and Addition of Matrices is listed below:

- $A + B = B + A$ (Commutative)
- $(A + B) + C = A + (B + C)$ (Associative)
- $AB \neq BA$ (Not Commutative)
- $(AB) C = A (BC)$ (Associative)
- $A (B+C) = AB + AC$ (Distributive)

Homogenous Coordinates

- To perform a sequence of transformation such as translation followed by rotation and scaling, we need to follow a sequential process −

• Translate the coordinates,

• Rotate the translated coordinates, and then

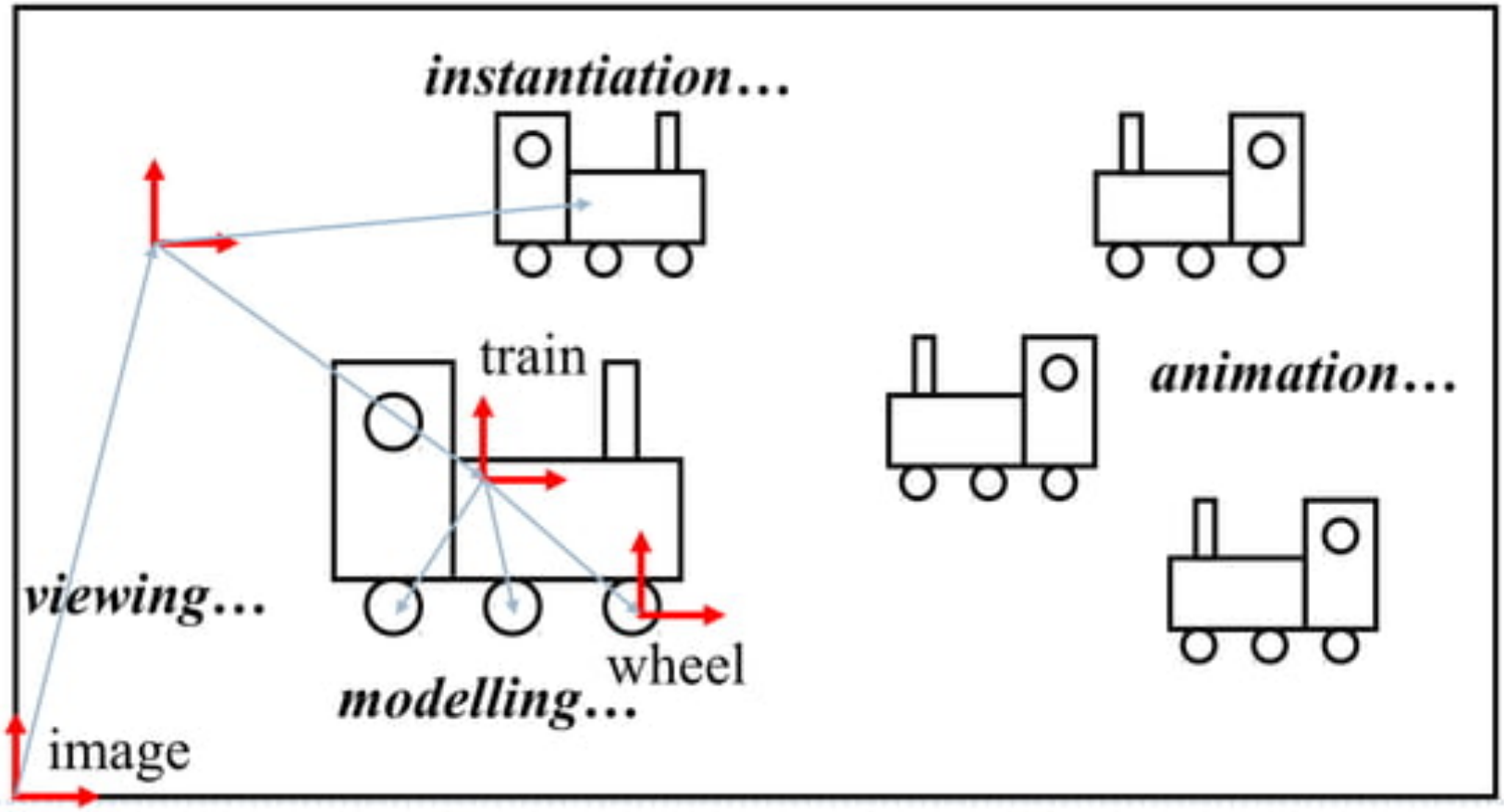
• Scale the rotated coordinates to complete the composite transformation.

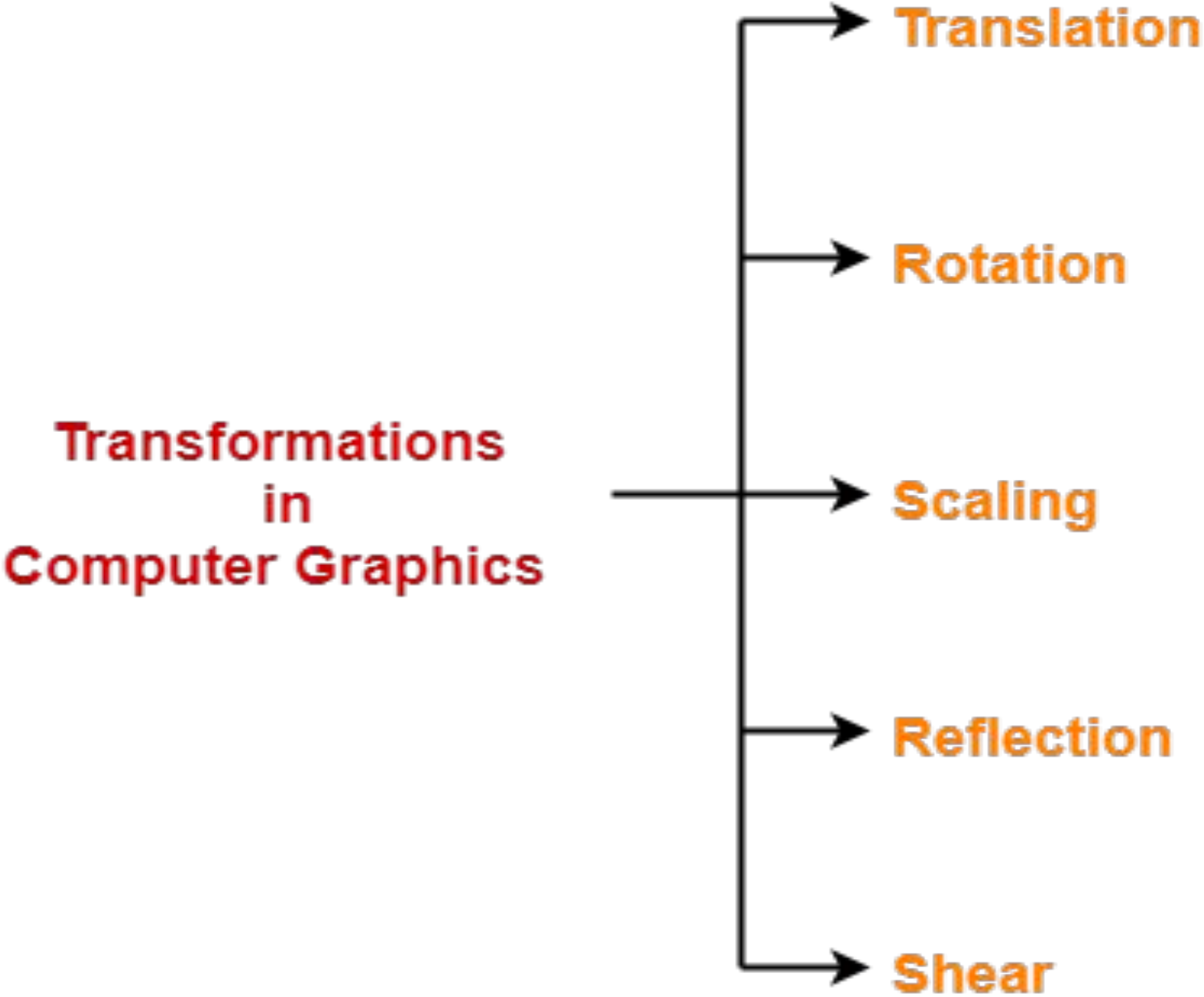
- To shorten this process, we have to use 3×3 transformation matrix instead of 2×2 transformation matrix.
- To convert a 2×2 matrix to 3×3 matrix, we have to add an extra dummy coordinate W.

- In this way, we can represent the point by 3 numbers instead of 2 numbers, which is called **Homogenous Coordinate system**.
- In this system, we can represent all the transformation equations in **matrix multiplication**.
- Any Cartesian point **P(X, Y)** can be converted to homogenous coordinates by **P' (Xh, Yh, h).**

Transformation

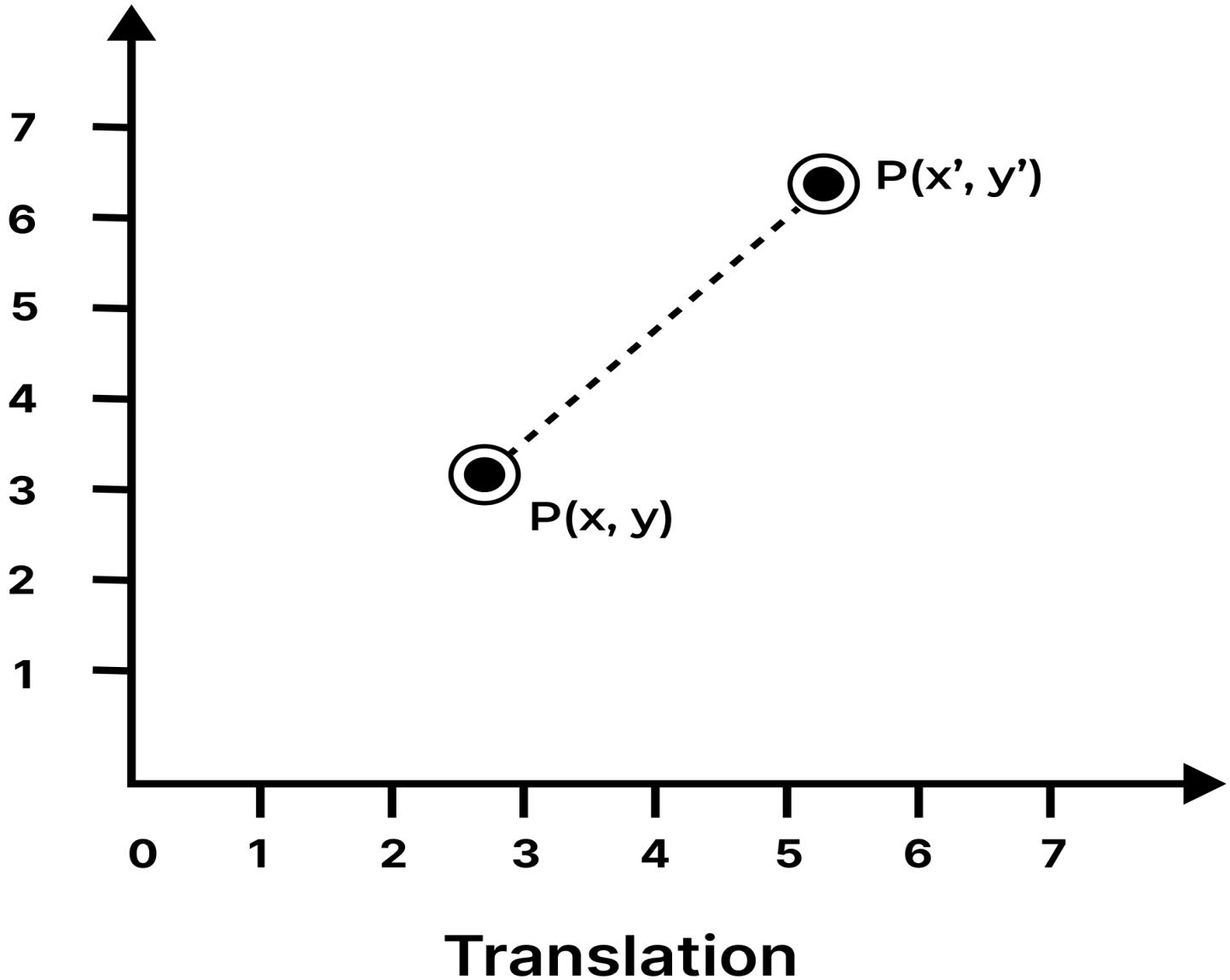
- Transformation changes the way object appears.
- Implementing changes in size of object, its position on screen or its orientation called Transformation.



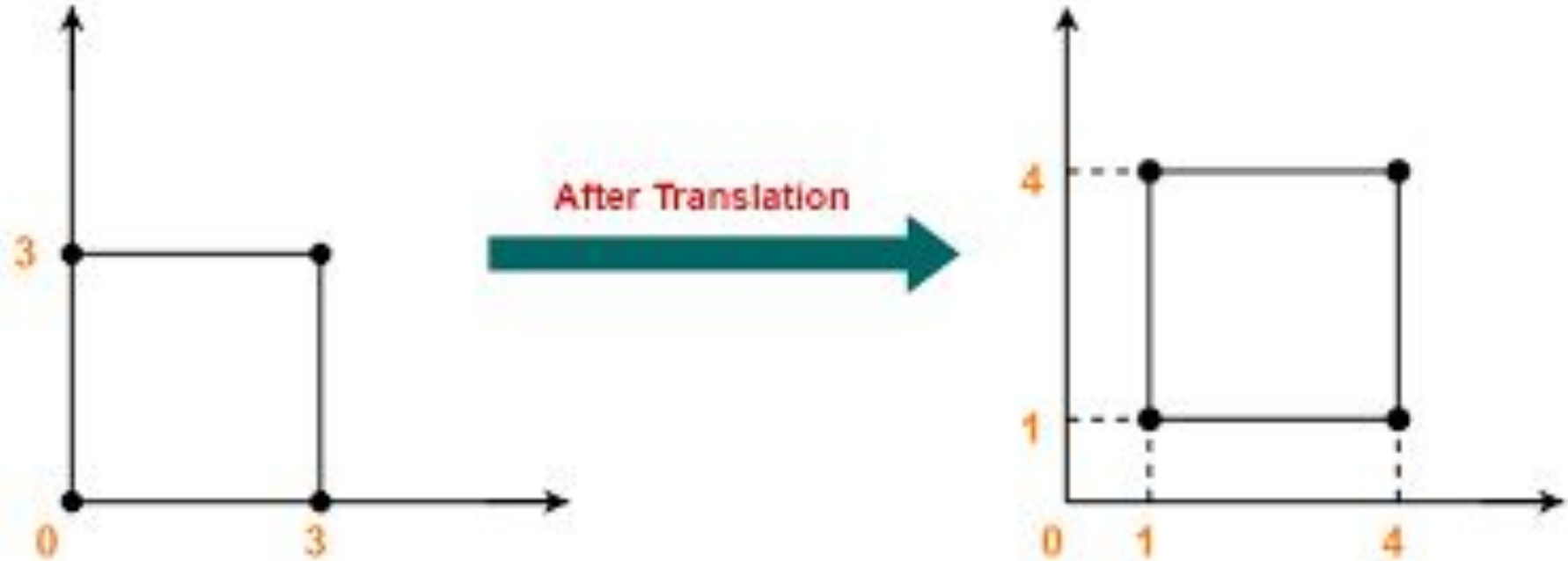


https://medium.com/data-science/understanding-transformations-in-computer-vision-b001f49a9e61

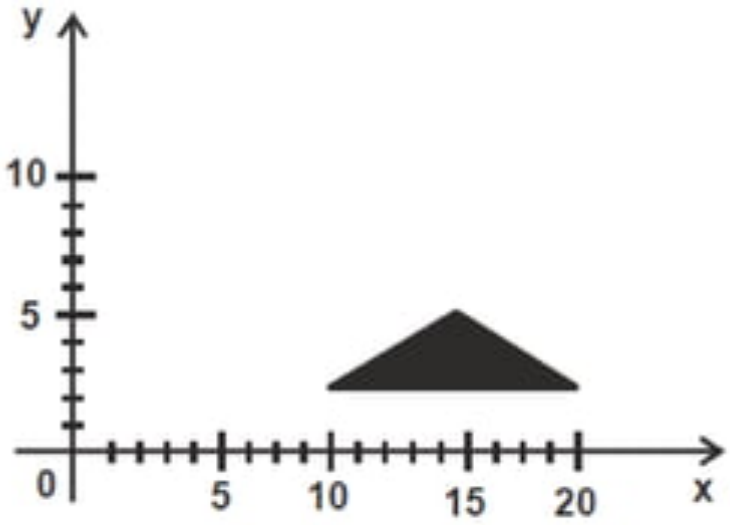
1) Translation

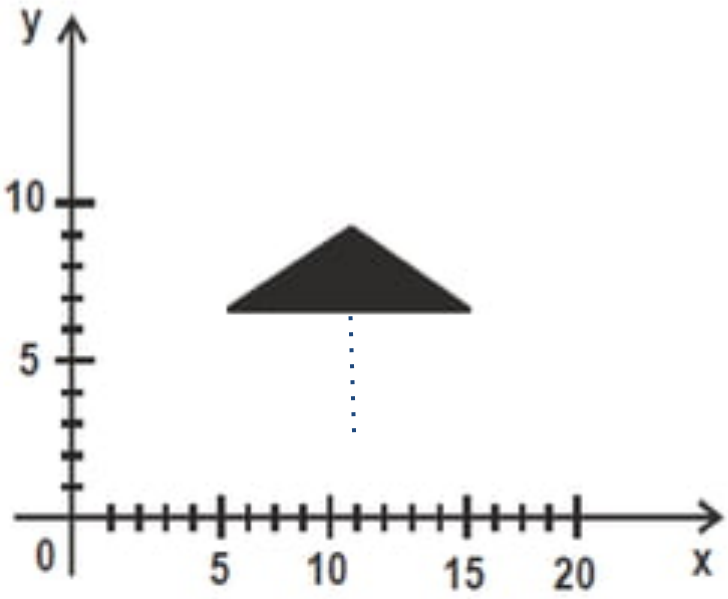


1) Translation



Translation





2D Translation

- In Computer graphics, 2D Translation is a **process of moving an object from one position to another in a two dimensional plane**.
- Consider a point object O has to be moved from one position to another in a 2D plane.
- Let-

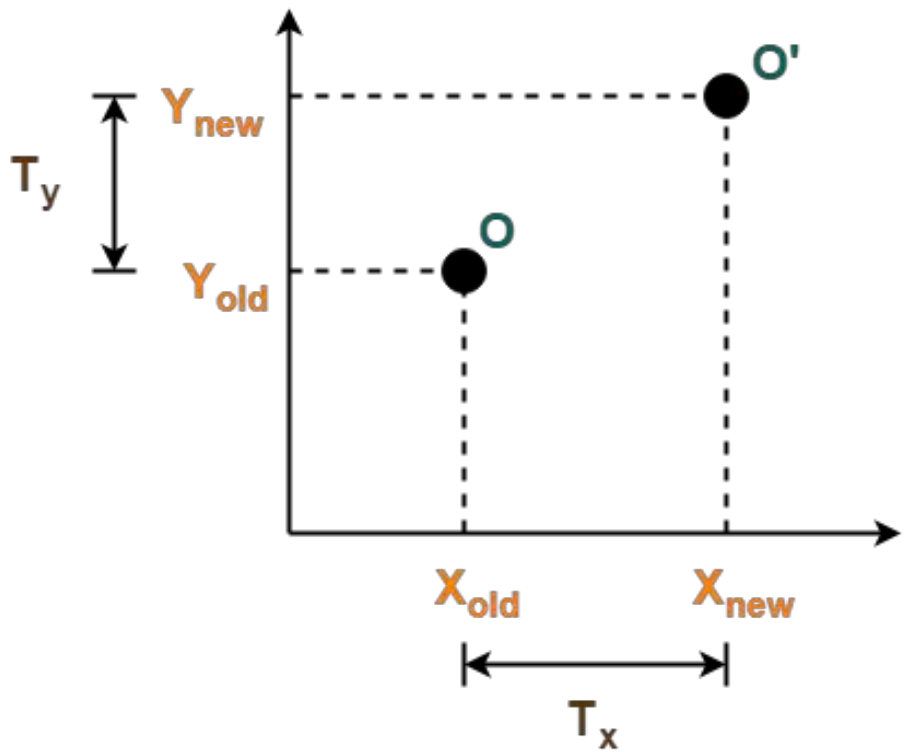
- Initial coordinates of the object O = **(Xold, Yold)**

- New coordinates of the object O after translation = **(Xnew, Ynew)**

- Translation vector or Shift vector = **(Tx, Ty)**

- Given a Translation vector (Tx, Ty)-

- Tx defines the distance the Xold coordinate has to be moved.
- Ty defines the distance the Yold coordinate has to be moved.



2D Translation in Computer Graphics

- This translation is achieved by adding the translation coordinates to the old coordinates of the object as-

$$X_{new} = X_{old} + T_x$$ (This denotes translation towards X axis)

$$Y_{new} = Y_{old} + T_y$$ (This denotes translation towards Y axis)

- In Matrix form, the above translation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix}$$

Translation Matrix

- The **homogeneous coordinates** representation of (X, Y) is (X, Y, 1).
- Through this representation, all the transformations can be performed using matrix / vector multiplications.
- The above translation matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Translation Matrix

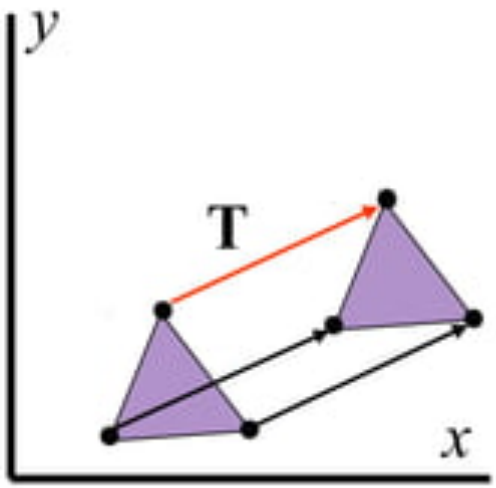
(Homogeneous Coordinates Representation)

Translation polygon

Translate polygon:

Apply the same operation on all points.

Works always, for all transformations of objects defined as a set of points.



Problem-01:

Given a circle C with radius 10 and center coordinates (1, 4). Apply the translation with distance 5 towards X axis and 1 towards Y axis. Obtain the new coordinates of C without changing its radius.

Solution-

Given-

- **Old center coordinates of C = (X_{old}, Y_{old}) = (1, 4)**
- **Translation vector = (T_x, T_y) = (5, 1)**

Let the new center coordinates of C = (X_{new}, Y_{new}).

Applying the translation equations, we have-

- $X_{new} = X_{old} + T_x = 1 + 5 = 6$
- $Y_{new} = Y_{old} + T_y = 4 + 1 = 5$

Thus, New center coordinates of C = (6, 5).

Alternatively,

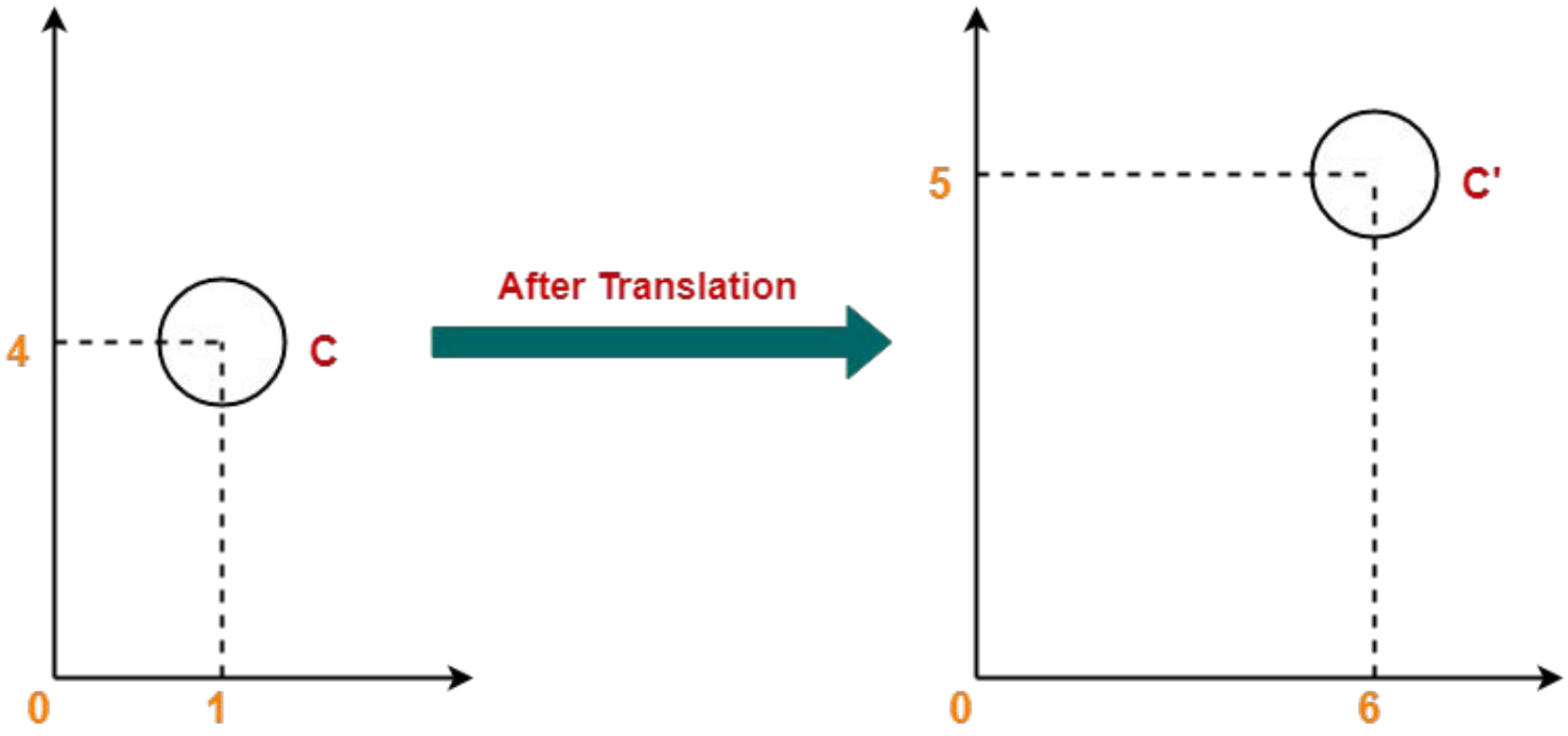
In matrix form, the new center coordinates of C after translation may be obtained as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix}$$

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

Thus, New center coordinates of C = (6, 5).


4
C
0
1
After Translation
5
C'
0
6

Problem 2)

Given a square with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the translation with distance 1 towards X axis and 1 towards Y axis. Obtain the new coordinates of the square.

<u>Solution-</u>

Given-

Old coordinates of the square = A (0, 3), B(3, 3), C(3, 0), D(0, 0)

Translation vector = $(T_x, T_y) = (1, 1)$

1) <u>For Coordinates A(0, 3)</u>

Let the new coordinates of corner A = (X_{new}, Y_{new}).

Applying the translation equations, we have-

$X_{new} = X_{old} + T_x = 0 + 1 = 1$

$Y_{new} = Y_{old} + T_y = 3 + 1 = 4$

Thus, New coordinates of corner A = (1, 4).

2) For Coordinates B(3, 3)

Let the new coordinates of corner B = (Xnew, Ynew).

Applying the translation equations, we have-

Xnew = Xold + Tx = 3 + 1 = 4
Ynew = Yold + Ty = 3 + 1 = 4
Thus, New coordinates of corner B = (4, 4).

3) For Coordinates C(3, 0)

Let the new coordinates of corner C = (X_{new}, Y_{new}).

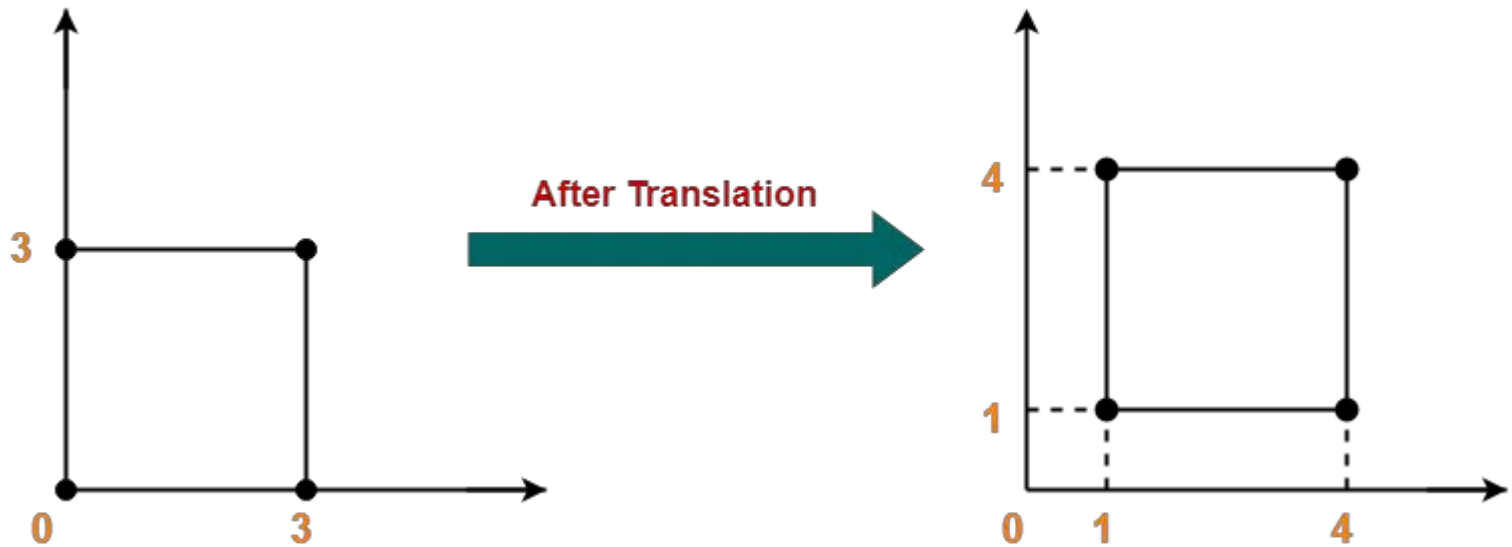
Applying the translation equations, we have-
$X_{new} = X_{old} + T_x = 3 + 1 = 4$
$Y_{new} = Y_{old} + T_y = 0 + 1 = 1$

Thus, New coordinates of corner C = (4, 1).

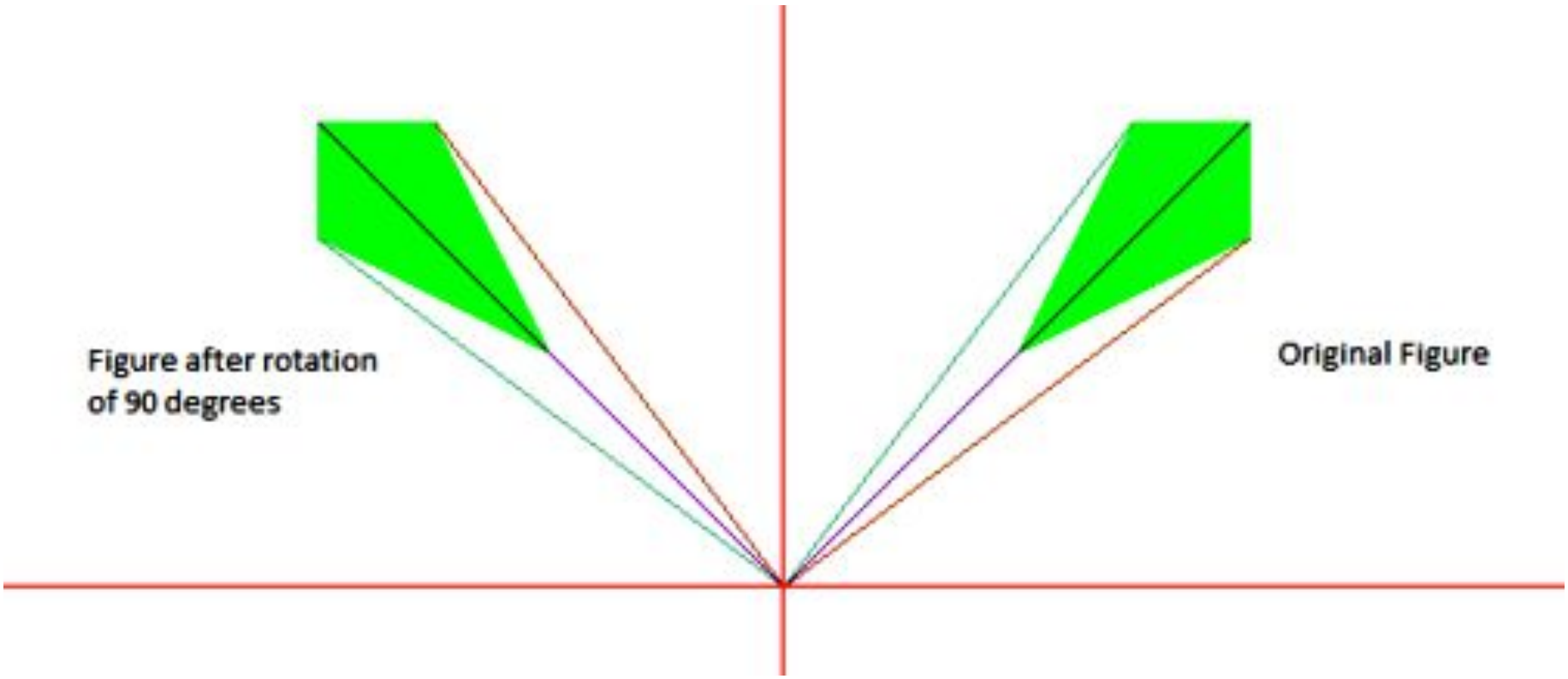
4) For Coordinates D(0, 0)

Let the new coordinates of corner D = (Xnew, Ynew).
Applying the translation equations, we have-

Xnew = Xold + Tx = 0 + 1 = 1
Ynew = Yold + Ty = 0 + 1 = 1
Thus, New coordinates of corner D = (1, 1).
So final Output:
New coordinates of the square = A (1, 4), B(4, 4), C(4, 1), D(1, 1).

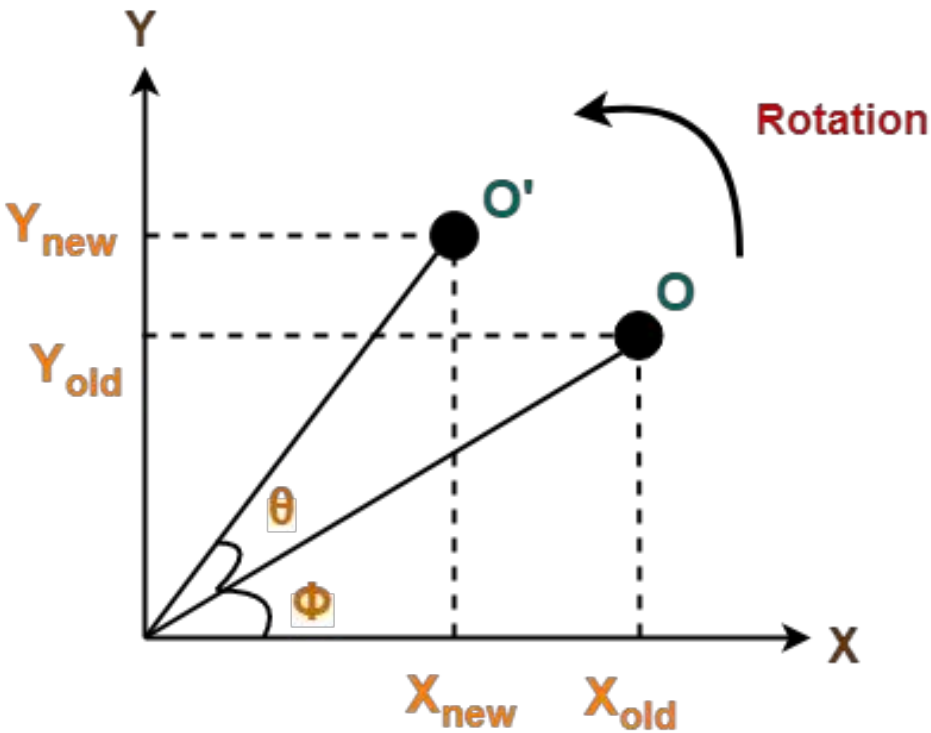


2D Rotation



2D Rotation

- In Computer graphics,2D Rotation is a process of rotating an object with respect to an angle in a two dimensional plane.



2D Rotation in Computer Graphics

- Consider a point object O has to be rotated from one angle to another in a 2D plane.

Let-

- Initial coordinates of the object O = (X_{old}, Y_{old})
- Initial angle of the object O with respect to origin = Φ
- Rotation angle = θ
- New coordinates of the object O after rotation = (X_{new}, Y_{new})

- **This rotation is achieved by using the following rotation equations-**

- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$
- $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$

- In Matrix form, the above rotation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Rotation Matrix

- For **homogeneous coordinates**, the above rotation matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Rotation Matrix

(Homogeneous Coordinates Representation)

PRACTICE PROBLEMS BASED ON 2D ROTATION IN COMPUTER GRAPHICS-

- **Problem-01:**
- Given a line segment with starting point as (0, 0) and ending point as (4, 4). Apply 30 degree rotation anticlockwise direction on the line segment and find out the new coordinates of the line.
- **Solution-**
- We rotate a straight line by its end points with the same angle. Then, we re-draw a line between the new end points.

- **Given-**
  - Old ending coordinates of the line = $(X_{old}, Y_{old}) = (4, 4)$
  - Rotation angle = $\theta = 30º$
- Let new ending coordinates of the line after rotation = (X_{new}, Y_{new}).
- Applying the rotation equations, we have-
- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$

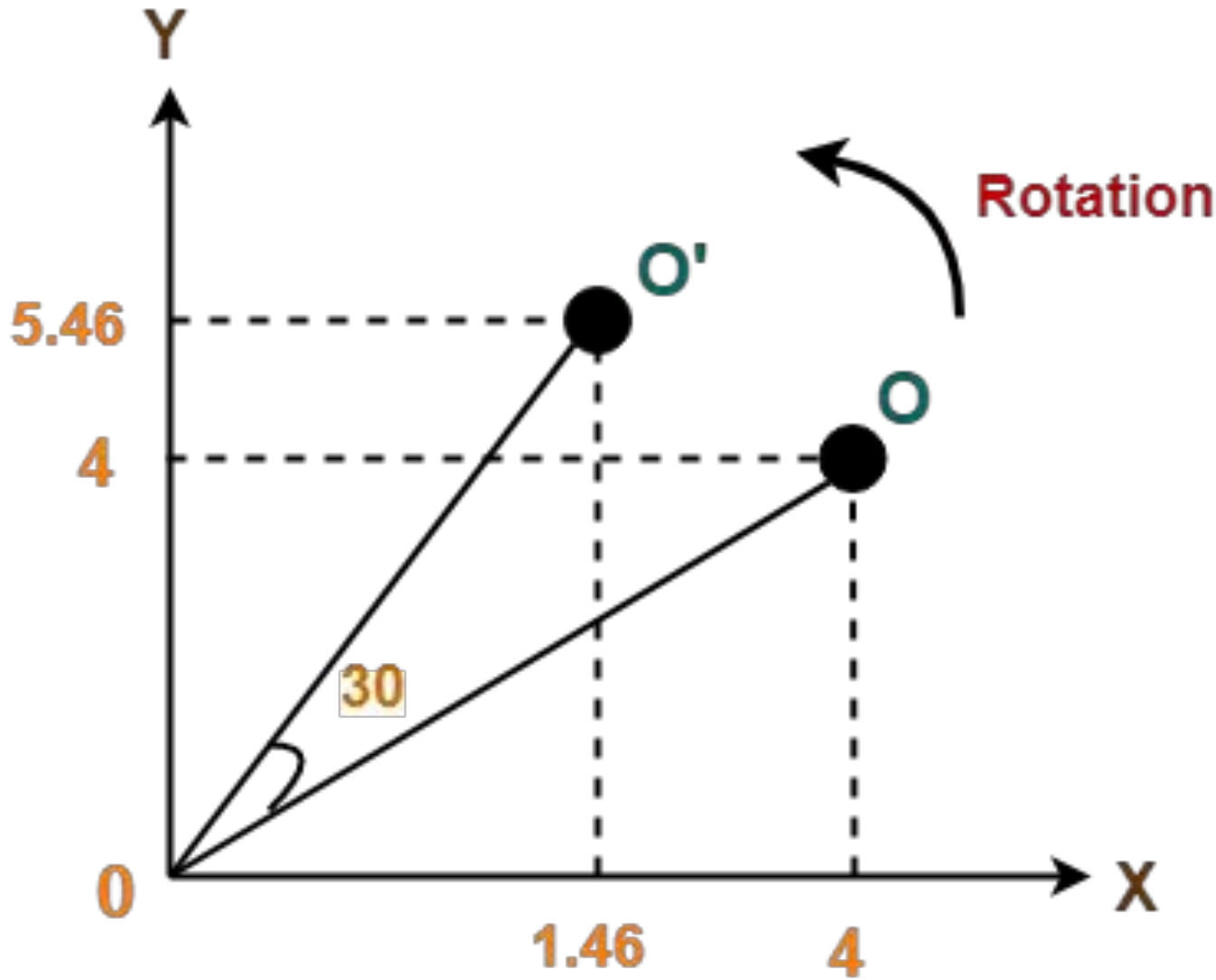
  $= 4 \times \cos 30º - 4 \times \sin 30º$

  $= 1.46$
- $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$

  $= 4 \times \sin 30º + 4 \times \cos 30º$

  $= 5.46$

Thus, New ending coordinates of the line after rotation = (1.46, 5.46).


Y
Rotation
O'
5.46
O
4
30
0
X
1.46
4

- **<u>Problem-02:</u>**
- **Given a triangle with corner coordinates (0, 0), (1, 0) and (1, 1). Rotate the triangle by 90 degree anticlockwise direction and find out the new coordinates.**
- **<u>Solution-</u>**
- We rotate a polygon by rotating each vertex of it with the same rotation angle.
- **Given-**
  - Old corner coordinates of the triangle = A (0, 0), B(1, 0), C(1, 1)
  - Rotation angle = $\theta = 90^{o}$

1)For Coordinates A(0, 0)

- Let the new coordinates of corner A after rotation = (X_{new}, Y_{new}).
- Applying the rotation equations, we have-
- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$

  $= 0 \times \cos 90^{o} - 0 \times \sin 90^{o}$

  $= 0$
- $Ynew = Xold \times \sin\theta + Yold \times \cos\theta$

  $= 0 \times \sin 90^{o} + 0 \times \cos 90^{o}$

  $= 0$
- **Thus, New coordinates of corner A after rotation = (0, 0).**

- **<u>For Coordinates B(1, 0)</u>**
- Let the new coordinates of corner B after rotation = (X_{new}, Y_{new}).
- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$

  $= 1 \times \cos 90º - 0 \times \sin 90º$

  $= 0$
- $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$

  $= 1 \times \sin 90º + 0 \times \cos 90º$

  $= 1 + 0$

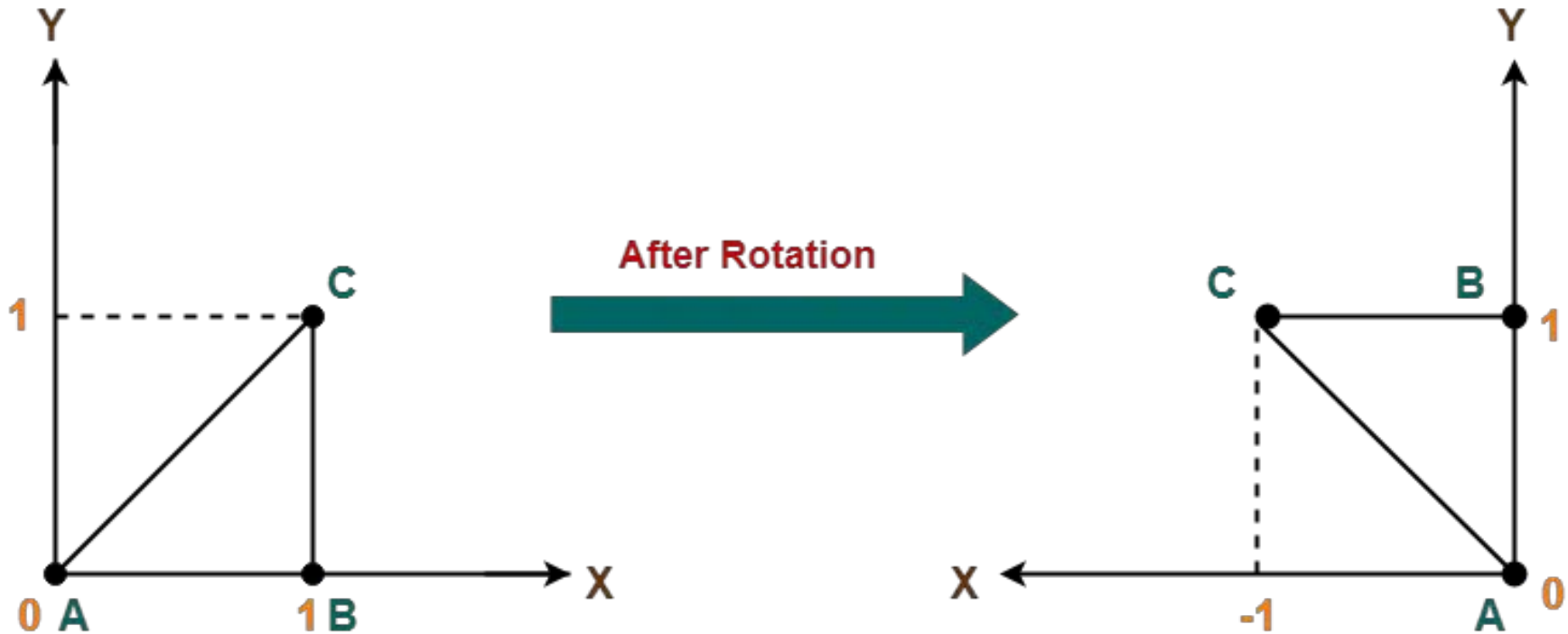
  $= 1$

Thus, New coordinates of corner B after rotation = (0, 1).

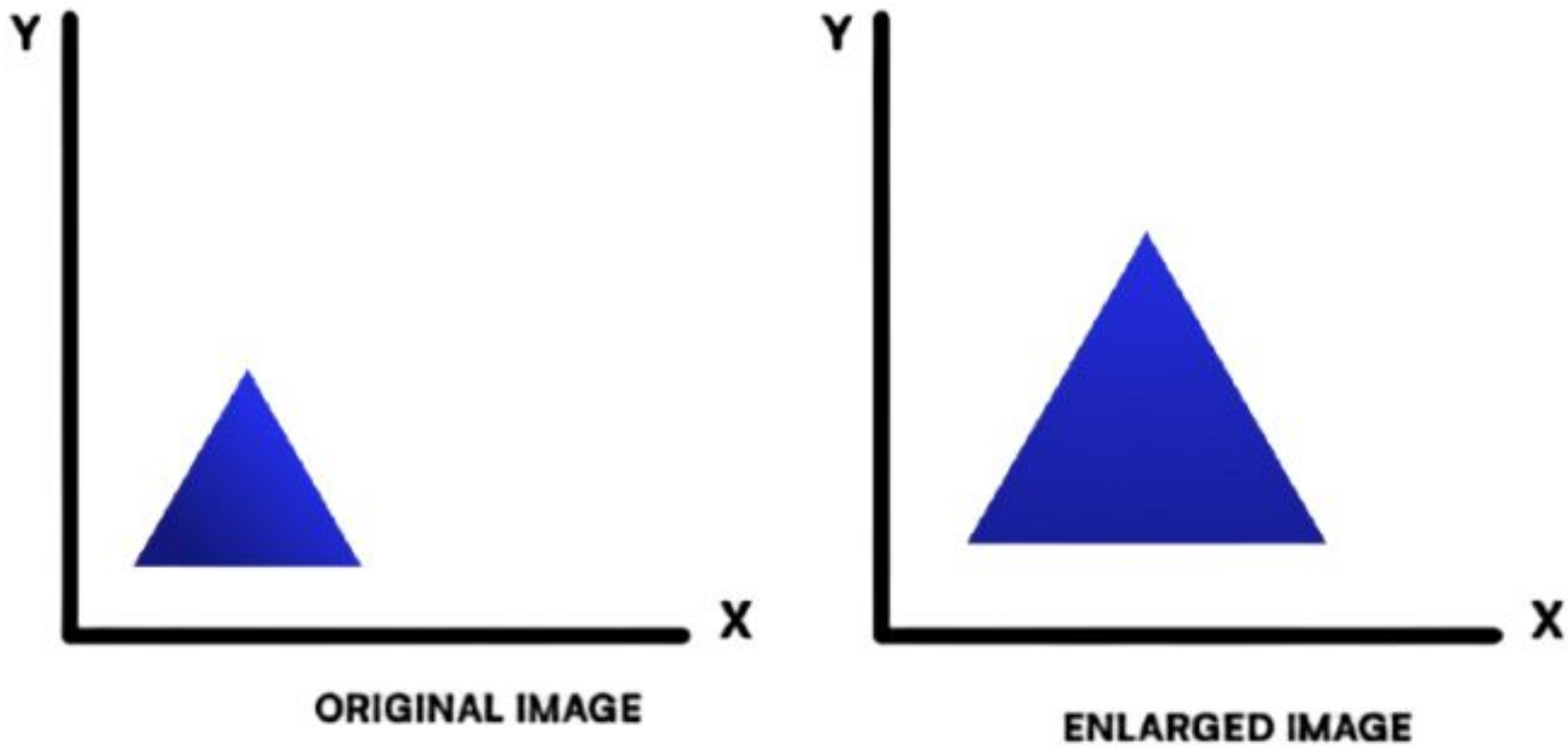
- **<u>For Coordinates C(1, 1)</u>**
- Let the new coordinates of corner C after rotation = (X_{new}, Y_{new}).

- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$
  $= 1 \times \cos 90º - 1 \times \sin 90º$
  $= 0 - 1$
  $= -1$
- $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$
  $= 1 \times \sin 90º + 1 \times \cos 90º$
  $= 1 + 0$
  $= 1$
- **Thus, New coordinates of corner C after rotation = (-1, 1).**

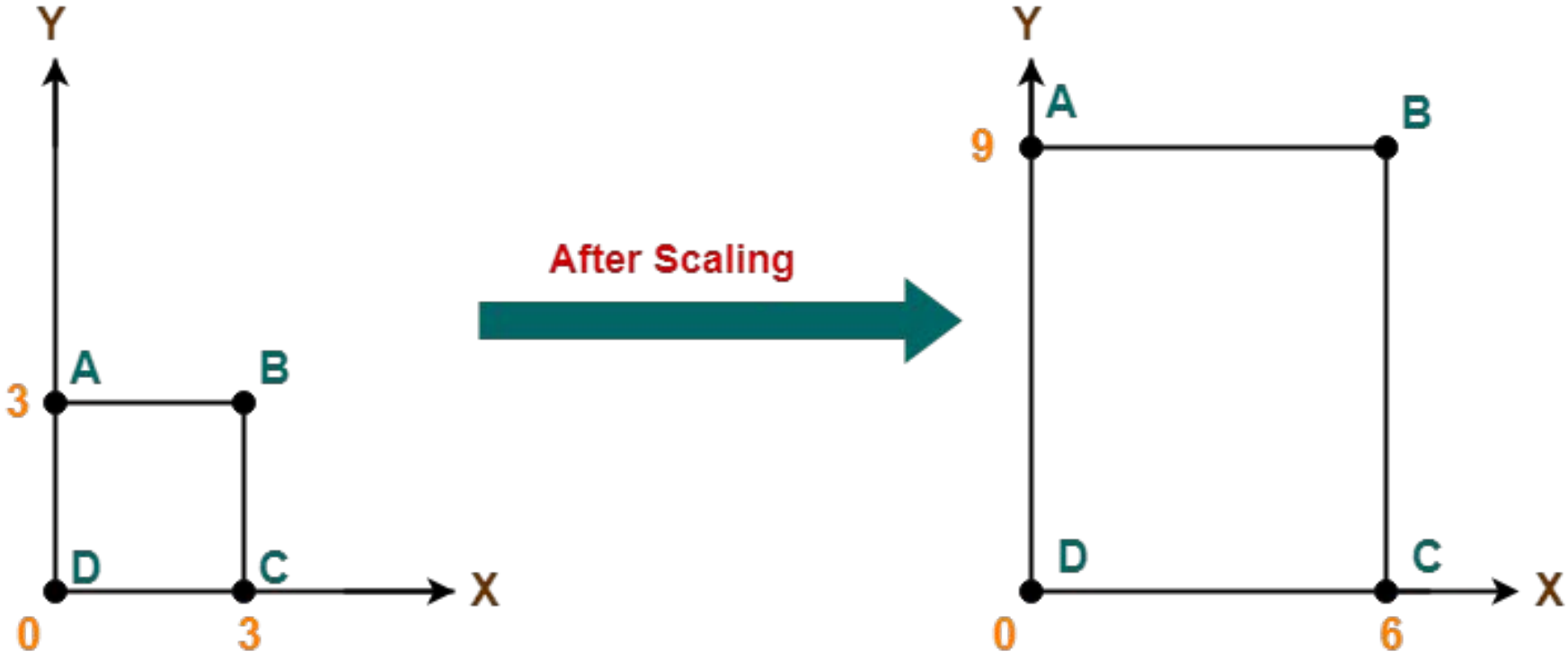
- **Thus, New coordinates of the triangle after rotation = A (0, 0), B(0, 1), C(-1, 1).**


Y
1
C
0 A
1 B
X
After Rotation
Y
C
B
1
X
-1
A
0

2D Scaling



2D Scaling



2D Scaling

- In computer graphics, scaling is a process of modifying or altering the size of objects.
- Scaling may be used to increase or reduce the size of object.
- Scaling subjects the coordinate points of the original object to change.
- Scaling factor determines whether the object size is to be increased or reduced.
- If **scaling factor > 1**, then the object **size is increased**.
- If **scaling factor < 1**, then the object **size is reduced.**

- Consider a point object O has to be scaled in a 2D plane.
- Let-
  - Initial coordinates of the object O = (X_{old}, Y_{old})
  - Scaling factor for X-axis = S_x
  - Scaling factor for Y-axis = S_y
  - New coordinates of the object O after scaling = (X_{new}, Y_{new})

- This scaling is achieved by using the following scaling equations-
- $X_{new} = X_{old} \times S_x$
- $Y_{new} = Y_{old} \times S_y$

- In Matrix form, the above scaling equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Scaling Matrix

For **homogeneous coordinates**, the above scaling matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Scaling Matrix

(Homogeneous Coordinates Representation)

PRACTICE PROBLEMS BASED ON 2D SCALING IN COMPUTER GRAPHICS-

- **Problem-01:**
- **Given a square object with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the scaling parameter 2 towards X axis and 3 towards Y axis and obtain the new coordinates of the object.**
- **Solution-**
- **Given-**
- Old corner coordinates of the square = A (0, 3), B(3, 3), C(3, 0), D(0, 0)
- Scaling factor along X axis = 2
- Scaling factor along Y axis = 3

- **<u>For Coordinates A(0, 3)</u>**
- Let the new coordinates of corner A after scaling = (X_{new}, Y_{new}).
- Applying the scaling equations, we have-

$$Xnew = Xold \times Sx = 0 \times 2 = 0$$

$$Ynew = Yold \times Sy = 3 \times 3 = 9$$

Thus, New coordinates of corner A after scaling = (0, 9).

- **<u>For Coordinates B(3, 3)</u>**
- Let the new coordinates of corner B after scaling = (Xnew, Ynew).
- Applying the scaling equations, we have-

$$Xnew = Xold \times Sx = 3 \times 2 = 6$$

$$Ynew = Yold \times Sy = 3 \times 3 = 9$$

Thus, New coordinates of corner B after scaling = (6, 9).

- **<u>For Coordinates C(3, 0)</u>**
- Let the new coordinates of corner C after scaling = (X_{new}, Y_{new}).
- Applying the scaling equations, we have-

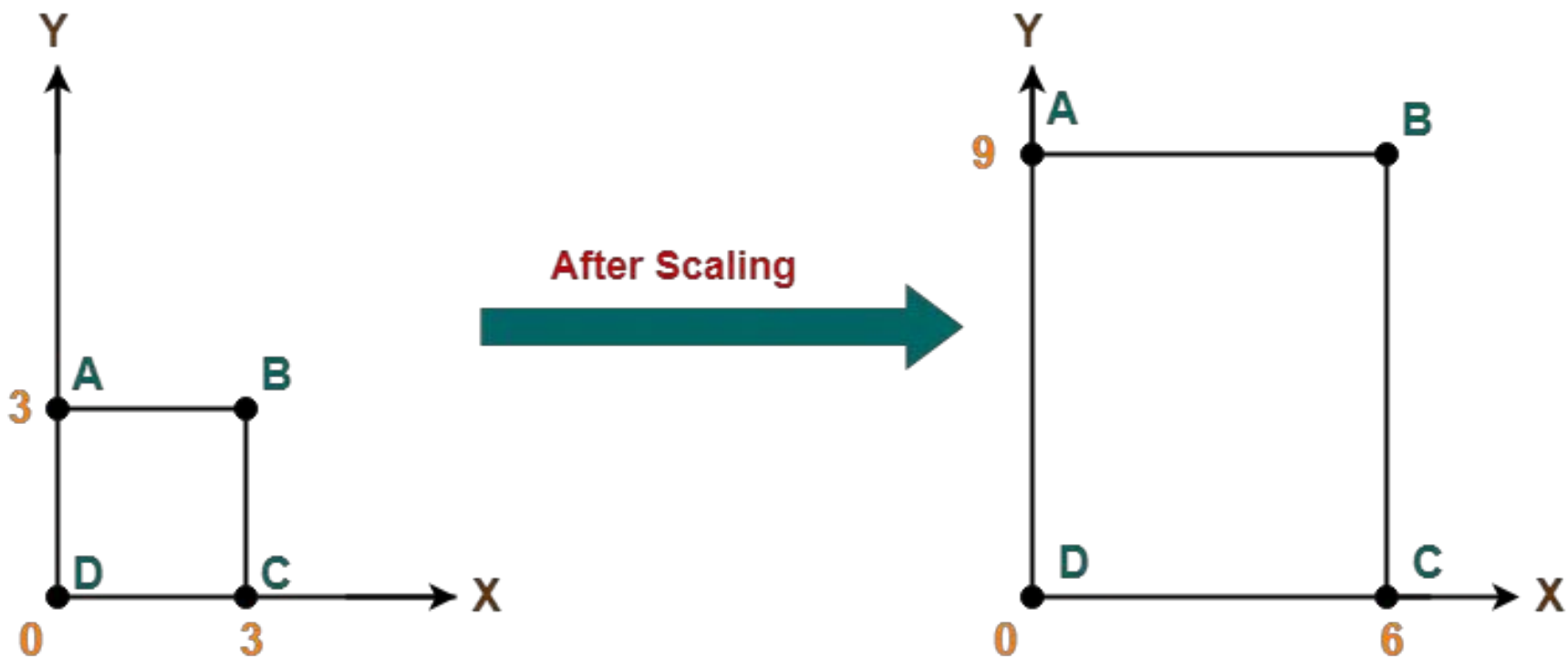
$$X_{new} = X_{old} \times S_x = 3 \times 2 = 6$$
$$Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$$

Thus, New coordinates of corner C after scaling = (6, 0).

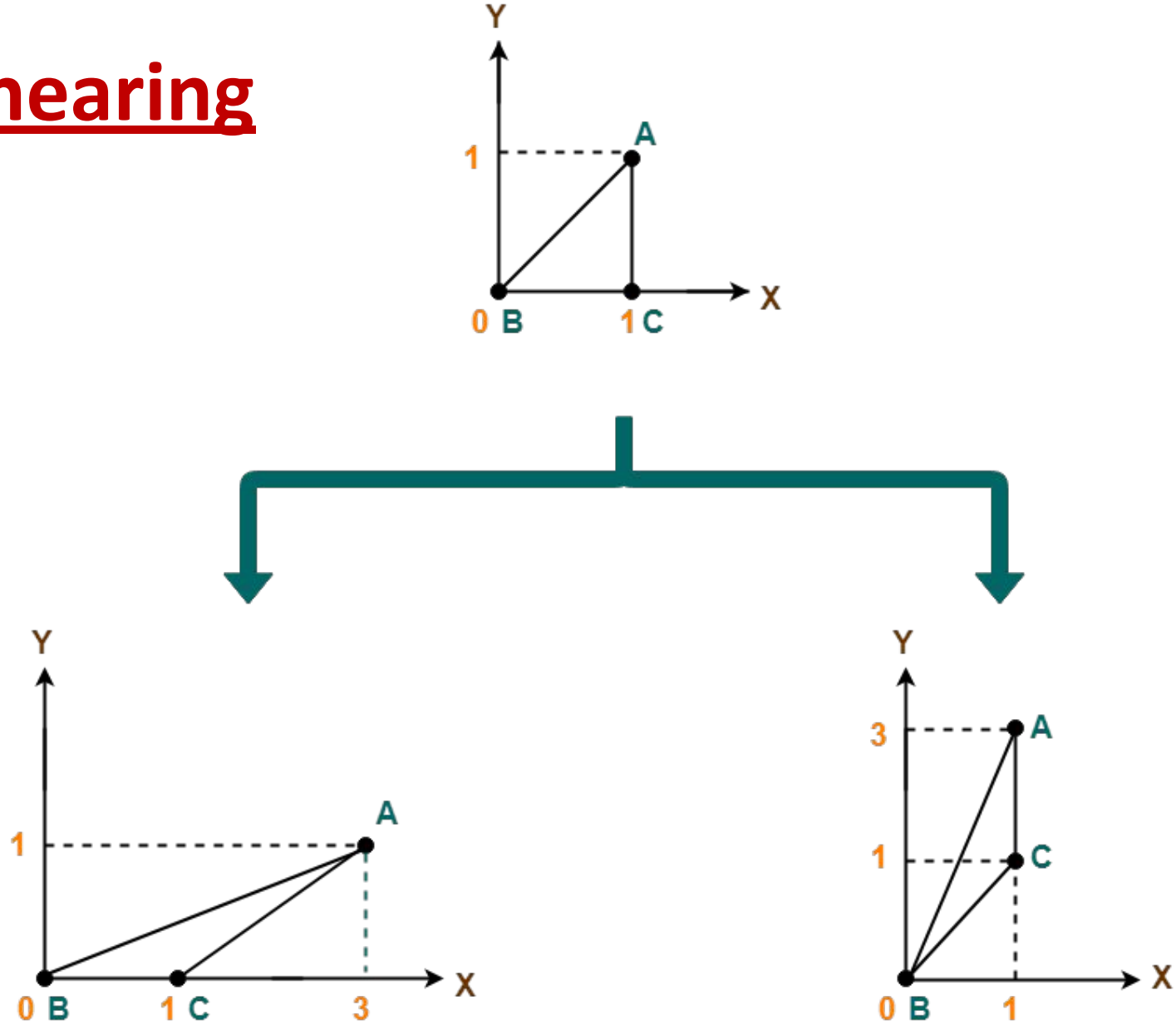
- **<u>For Coordinates D(0, 0)</u>**
- Let the new coordinates of corner D after scaling = (Xnew, Ynew).
- Applying the scaling equations, we have-

$$Xnew = Xold \times Sx = 0 \times 2 = 0$$
$$Ynew = Yold \times Sy = 0 \times 3 = 0$$

- **Thus, New coordinates of corner D after scaling = (0, 0).**
- **Thus, New coordinates of the square after scaling = A (0, 9), B(6, 9), C(6, 0), D(0, 0).**



2D Shearing

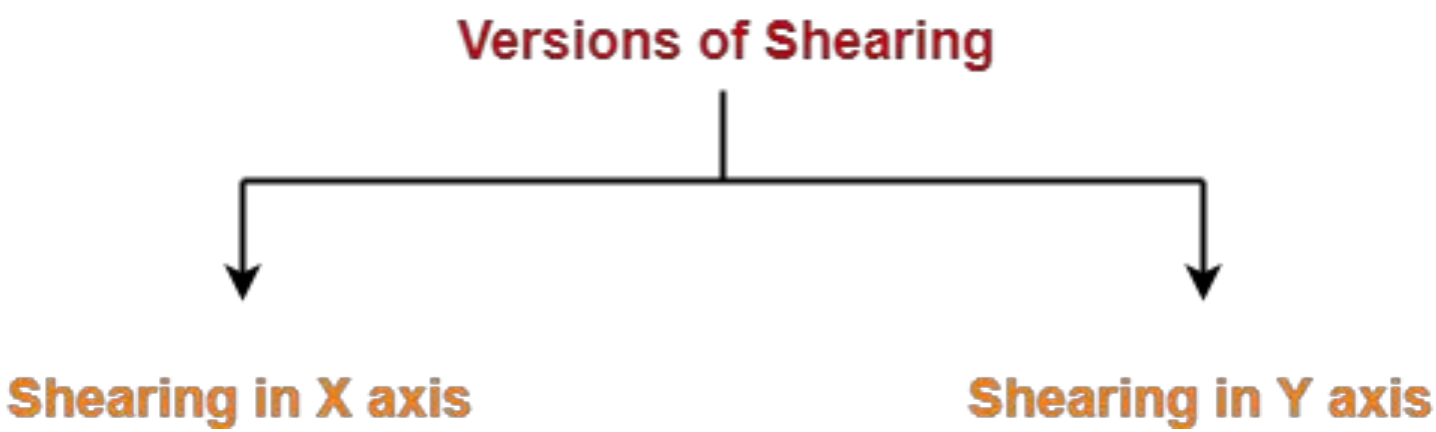


Shearing in X Axis

Shearing in Y Axis

2D Shearing

- In Computer graphics,2D Shearing is an ideal technique to change the shape of an existing object in a two dimensional plane.
- In a two dimensional plane, the object size can be changed along X direction as well as Y direction.
- So, there are two versions of shearing-



- Consider a point object O has to be sheared in a 2D plane.
- Let-
- Initial coordinates of the object O = $(\mathbf{X_{old}}, \mathbf{Y_{old}})$
- Shearing parameter towards X direction = $\mathbf{Sh_x}$
- Shearing parameter towards Y direction = $\mathbf{Sh_y}$
- New coordinates of the object O after shearing = $(\mathbf{X_{new}}, \mathbf{Y_{new}})$

<u>1) Shearing in X Axis-</u>

- Shearing in X axis is achieved by using the following shearing equations-

$$Xnew = Xold + Shx \times Yold$$

$$Ynew = Yold$$

In Matrix form, the above shearing equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 & Sh_x \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Shearing Matrix

(In X axis)

For **homogeneous coordinates**, the above shearing matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ Sh_x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Shearing Matrix

(In X axis)

(Homogeneous Coordinates Representation)

- **<u>Shearing in Y Axis-</u>**
- Shearing in Y axis is achieved by using the following shearing equations-

$$X_{new} = X_{old}$$
$$Y_{new} = Y_{old} + Sh_y \times X_{old}$$

- In Matrix form, the above shearing equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Sh_y & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Shearing Matrix

(In Y axis)

- For **homogeneous coordinates**, the above shearing matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Sh_y & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Shearing Matrix

(In Y axis)

(Homogeneous Coordinates Representation)

PRACTICE PROBLEMS BASED ON 2D SHEARING IN COMPUTER GRAPHICS-

- **Problem-01:**

- Given a triangle with points (1, 1), (0, 0) and (1, 0). Apply shear parameter 2 on X axis and 2 on Y axis and find out the new coordinates of the object.

- **Solution-**

- **Given-**
- Old corner coordinates of the triangle = A (1, 1), B(0, 0), C(1, 0)
- Shearing parameter towards X direction (Sh_x) = 2
- Shearing parameter towards Y direction (Sh_y) = 2

- **Shearing in X Axis-**
- **For Coordinates A(1, 1)**
- Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}).

- Applying the shearing equations, we have-

$$X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 1 = 3$$

$$Y_{new} = Y_{old} = 1$$

- **Thus, New coordinates of corner A after shearing = (3, 1).**
- **For Coordinates B(0, 0)**
- Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}).
- Applying the shearing equations, we have-

$$Xnew = Xold + Shx \times Yold = 0 + 2 \times 0 = 0$$

$$Ynew = Yold = 0$$

Thus, New coordinates of corner B after shearing = (0, 0).

- **<u>For Coordinates C(1, 0)</u>**
- Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}).

- Applying the shearing equations, we have-

$$X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 0 = 1$$

$$Y_{new} = Y_{old} = 0$$

- **Thus, New coordinates of corner C after shearing = (1, 0).**
- **Thus, New coordinates of the triangle after shearing in X axis = A (3, 1), B(0, 0), C(1, 0).**

- **Shearing in Y Axis-**
- **For Coordinates A(1, 1)**
- Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}).
- Applying the shearing equations, we have-

$$X_{new} = X_{old} = 1$$

$$Y_{new} = Y_{old} + Sh_y \times X_{old} = 1 + 2 \times 1 = 3$$

- **Thus, New coordinates of corner A after shearing = (1, 3).**
- **For Coordinates B(0, 0)**
- Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}).
- Applying the shearing equations, we have-

$X_{new} = X_{old} = 0$

$Y_{new} = Y_{old} + Sh_y \times X_{old} = 0 + 2 \times 0 = 0$

- **Thus, New coordinates of corner B after shearing = (0, 0).**

- **<u>For Coordinates C(1, 0)</u>**
- Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}).

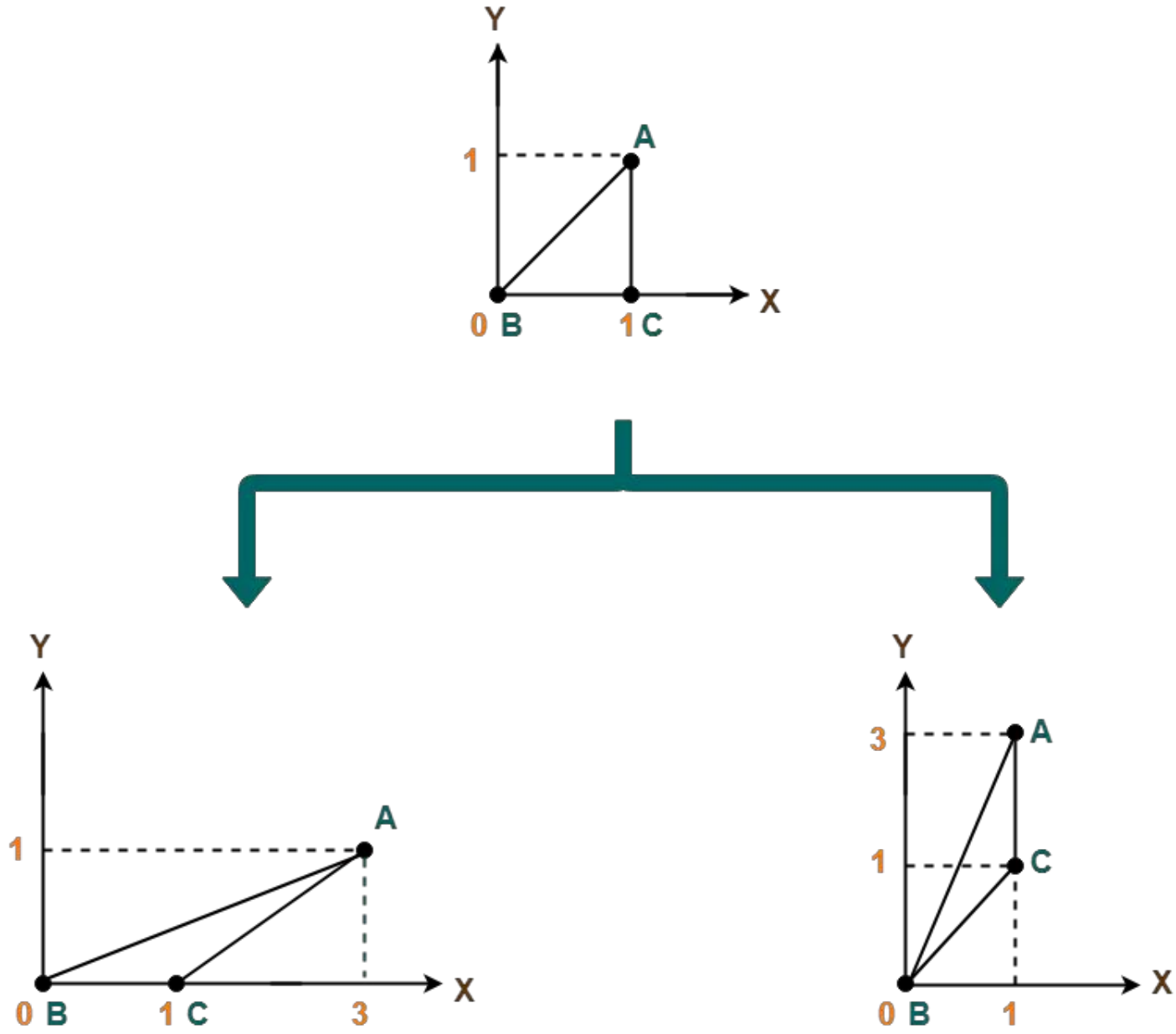
- Applying the shearing equations, we have-

$X_{new} = X_{old} = 1$

$Y_{new} = Y_{old} + Sh_y \times X_{old} = 0 + 2 \times 1 = 2$

- **Thus, New coordinates of corner C after shearing = (1, 2).**

- **Thus, New coordinates of the triangle after shearing in Y axis = A (1, 3), B(0, 0), C(1, 2).**



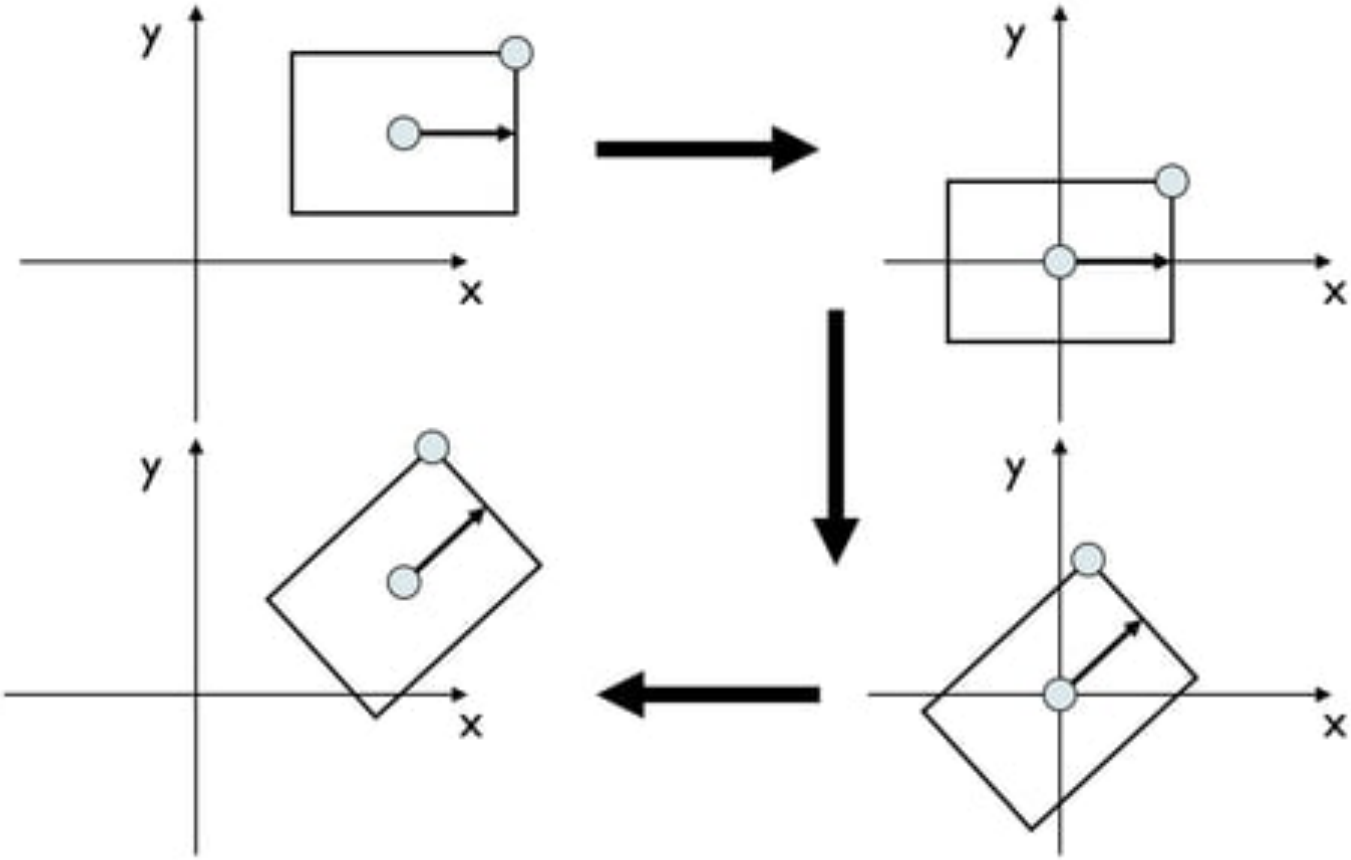
Shearing in X Axis

Shearing in Y Axis

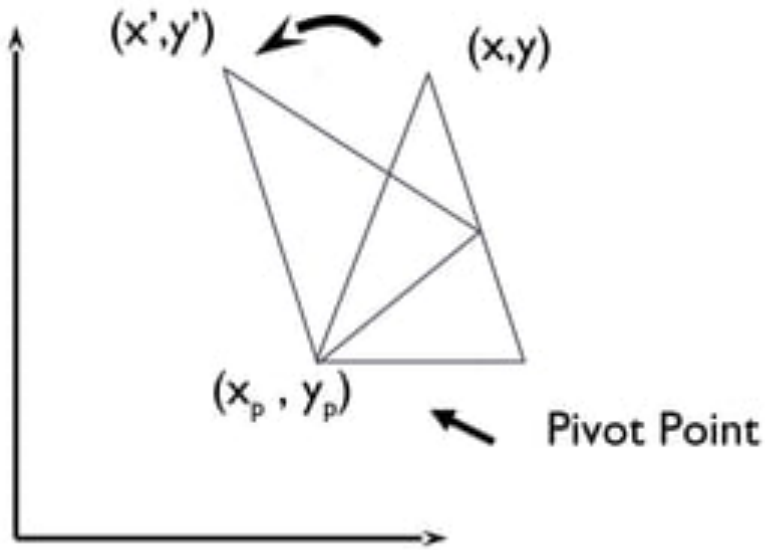
Rotating About An Arbitrary Point

- What happens when you apply a rotation transformation to an object that is not at the origin?
- Solution:
  - Translate the center of rotation to the origin
  - Rotate the object
  - Translate back to the original location

Rotating About An Arbitrary Point



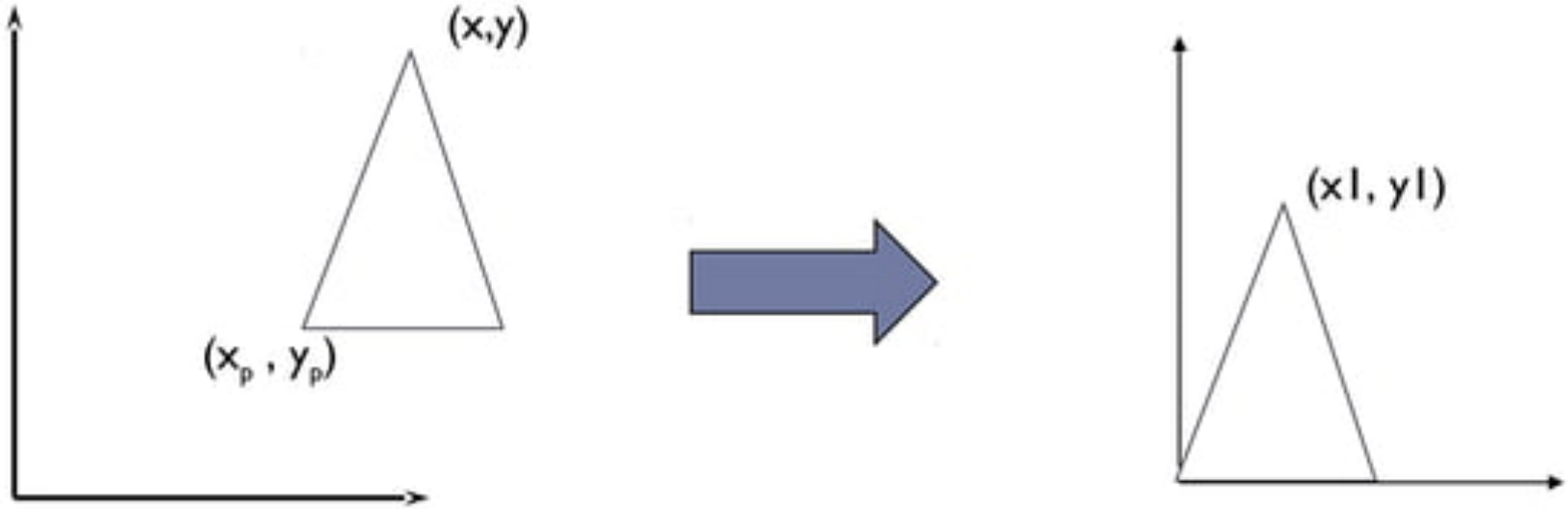
Rotation About a Pivot Point



- Pivot point is the point of rotation
- Pivot point need not necessarily be on the object

Rotation About a Pivot Point

STEP-1: Translate the pivot point to the origin

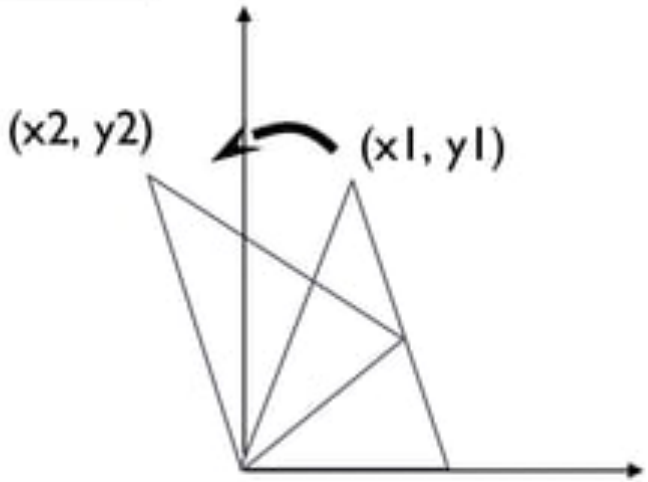


$$x1 = x - x_p$$
$$y1 = y - y_p$$

$$T1 = \begin{bmatrix} 1 & 0 & -xp \\ 0 & 1 & -yp \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation About a Pivot Point

STEP-2: **Rotate about the origin**

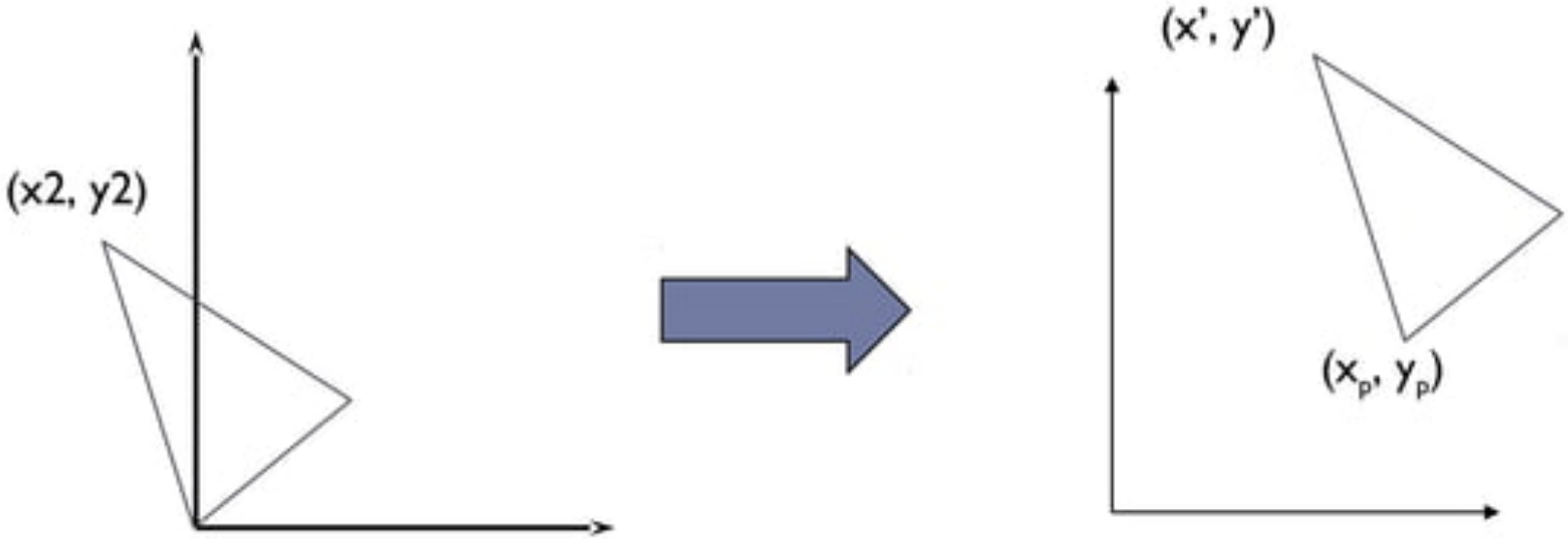


$$x2 = x1\cos\theta - y1\sin\theta$$
$$y2 = x1\sin\theta + y1\cos\theta$$

$$R = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation About a Pivot Point

STEP-3: **Translate the pivot point to original position**



$$x' = x2 + x_p$$

$$y' = y2 + y_p$$

$$T2 = \begin{bmatrix} 1 & 0 & xp \\ 0 & 1 & yp \\ 0 & 0 & 1 \end{bmatrix}$$

The Three-Step Process of Rotation about arbitrary point

1. **Translate to the Origin:** Perform an inverse translation of the arbitrary point (x_0, y_0) to move it to the origin (0,0). For every point (x, y) in your object or scene, you subtract (x_0, y_0) from it to get ($x - x_0$, $y - y_0$).

1. **Rotate About the Origin:** Now that the arbitrary point is at the origin, perform the actual rotation around (0,0) by the desired angle. The standard 2D rotation formulas are used here:
   a. **$x' = x \cos(\theta) - y \sin(\theta)$**
   b. **$y' = x \sin(\theta) + y \cos(\theta)$**
   c. Where (x, y) are the translated coordinates, θ is the rotation angle, and (x', y') are the new rotated coordinates.

3. Translate Back: Apply the inverse translation to move the arbitrary point from the origin back to its original position (x_0, y_0). This means you add (x_0, y_0) back to the rotated coordinates:

a. $x'' = x' + x_0$
b. $y'' = y' + y_0$
c. Where (x'', y'') are the final coordinates after the rotation about the arbitrary point.

Final Formula

$$x' = x_p + (x - x_p)\cos\theta - (y - y_p)\sin\theta$$

$$y' = y_p + (x - x_p)\sin\theta + (y - y_p)\cos\theta$$

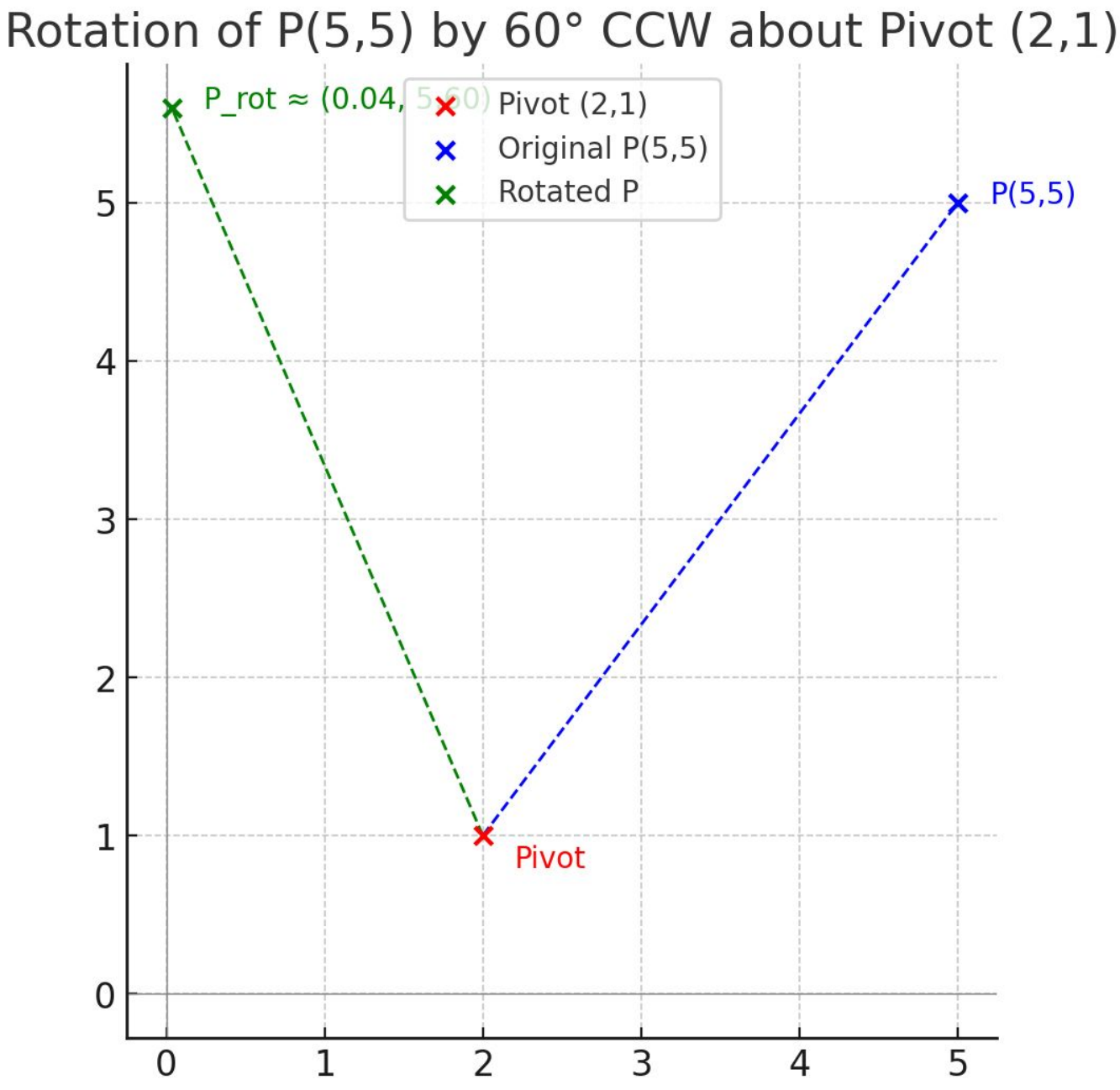
✅ This is the reusable formula for **any point, any pivot, any rotation angle.**

Problem 1) Rotate point P(5, 5) by an angle of 60 degrees counter-clockwise around the pivot point Pp(2, 1).

Solution:

- **Translate to Origin**: Subtract the pivot point's coordinates from P's coordinates.
  - $(x - xp) = 5 - 2 = 3$
  - $(y - yp) = 5 - 1 = 4$
  - So, the point relative to the origin is (3, 4).
- **Rotate about the Origin**: Apply the 2D rotation formula for a point rotated around the origin.
  - $\theta = 60$ degrees
  -
  - $\cos(60°) = 0.5$
  - $\sin(60°) = \sqrt{3}/2 \approx 0.866$
  - $x'' = (3 * 0.5) - (4 * 0.866) = 1.5 - 3.464 = -1.964$
  - $y'' = (3 * 0.866) + (4 * 0.5) = 2.598 + 2.0 = 4.598$
  - The rotated point relative to the origin is approximately (-1.964, 4.598).

- **Translate Back**: Add the pivot point's coordinates back to the result.
  - $x' = -1.964 + 2 = 0.036$
  - $y' = 4.598 + 1 = 5.598$
  - The final rotated point $P'(x', y')$ is approximately (0.036, 5.598).


Rotation of P(5,5) by 60° CCW about Pivot (2,1)
P_rot ≈ (0.04, 5.60)
Pivot (2,1)
Original P(5,5)
Rotated P
P(5,5)
Pivot
0
1
2
3
4
5
0
1
2
3
4
5

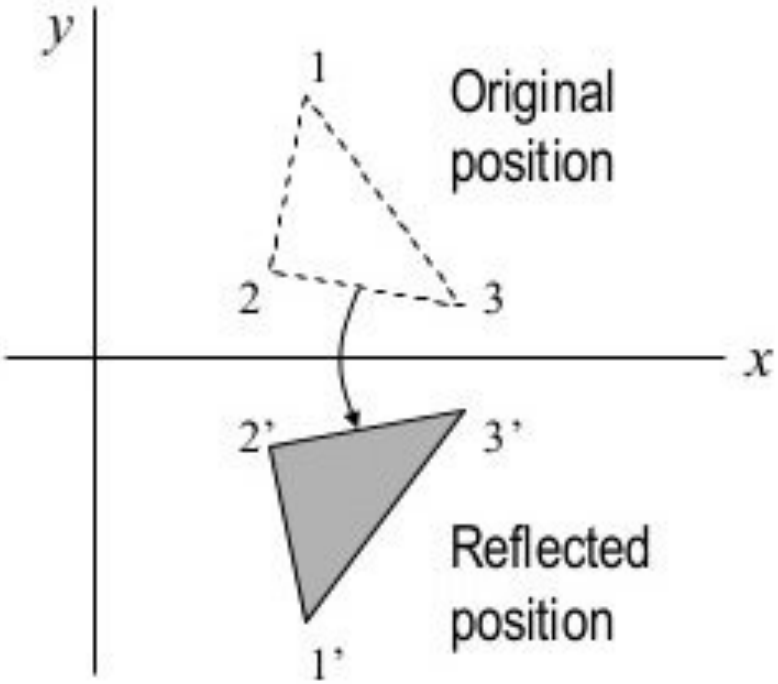
2D Reflection

- Reflection is a kind of rotation where the angle of **rotation is 180 degree**.

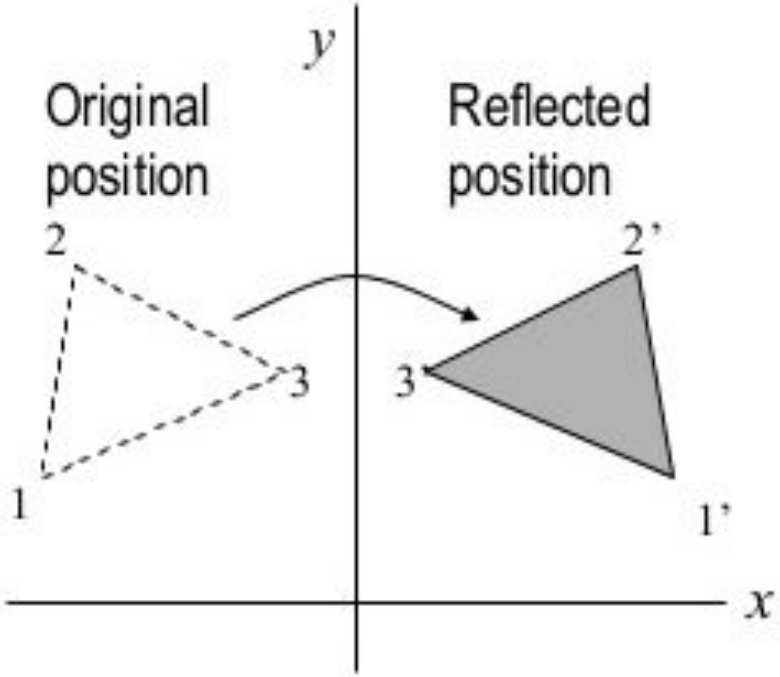
- The reflected object is always formed on the other side of **mirror**.

- The **size of reflected object is same** as the size of original object.

Reflection



Reflection about x-axis



Reflection about y-axis

- Consider a point object O has to be reflected in a 2D plane.
- Let-
- Initial coordinates of the object O = (X_{old}, Y_{old})
- New coordinates of the reflected object O after reflection = (X_{new}, Y_{new})

- **Reflection On X-Axis:**
- This reflection is achieved by using the following reflection equations-

$$X_{new} = X_{old}$$
$$Y_{new} = -Y_{old}$$

- In Matrix form, the above reflection equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Reflection Matrix

(Reflection Along X Axis)

- For homogeneous coordinates, the above reflection matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Reflection Matrix

(Reflection Along X Axis)

(Homogeneous Coordinates Representation)

- **Reflection On Y-Axis:**
- This reflection is achieved by using the following reflection equations-

$$X_{new} = -X_{old}$$
$$Y_{new} = Y_{old}$$

- In Matrix form, the above reflection equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Reflection Matrix

(Reflection Along Y Axis)

- For homogeneous coordinates, the above reflection matrix may be represented as a 3 x 3 matrix as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ 1 \end{bmatrix}$$

Reflection Matrix

(Reflection Along Y Axis)

(Homogeneous Coordinates Representation)

- **<u>Problem-01:</u>**

Given a triangle with coordinate points A(3, 4), B(6, 4), C(5, 6). Apply the reflection on the X axis and obtain the new coordinates of the object.

- **<u>Solution-</u>**
- Given-
- Old corner coordinates of the triangle = A (3, 4), B(6, 4), C(5, 6)
- Reflection has to be taken on the X axis
- **<u>For Coordinates A(3, 4)</u>**
- Let the new coordinates of corner A after reflection = (X_{new}, Y_{new}).
- Applying the reflection equations, we have-

$$\mathbf{X_{new} = X_{old} = 3}$$
$$\mathbf{Y_{new} = -Y_{old} = -4}$$

Thus, New coordinates of corner A after reflection = (3, -4).

- **<u>For Coordinates B(6, 4)</u>**
- Let the new coordinates of corner B after reflection = (X_{new}, Y_{new}).
- Applying the reflection equations, we have-

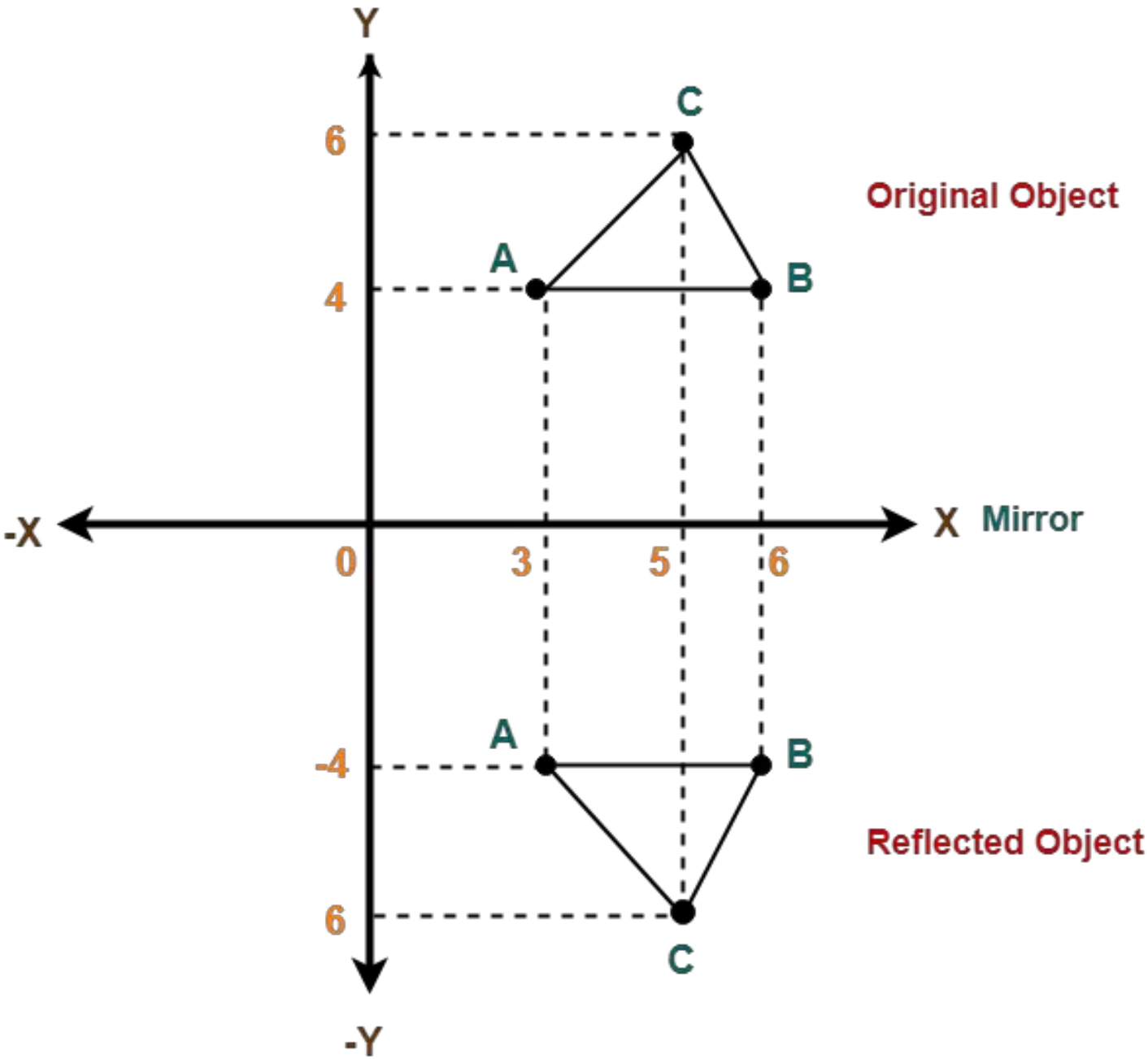
$$X_{new} = X_{old} = 6$$
$$Y_{new} = -Y_{old} = -4$$

Thus, New coordinates of corner B after reflection = (6, -4).

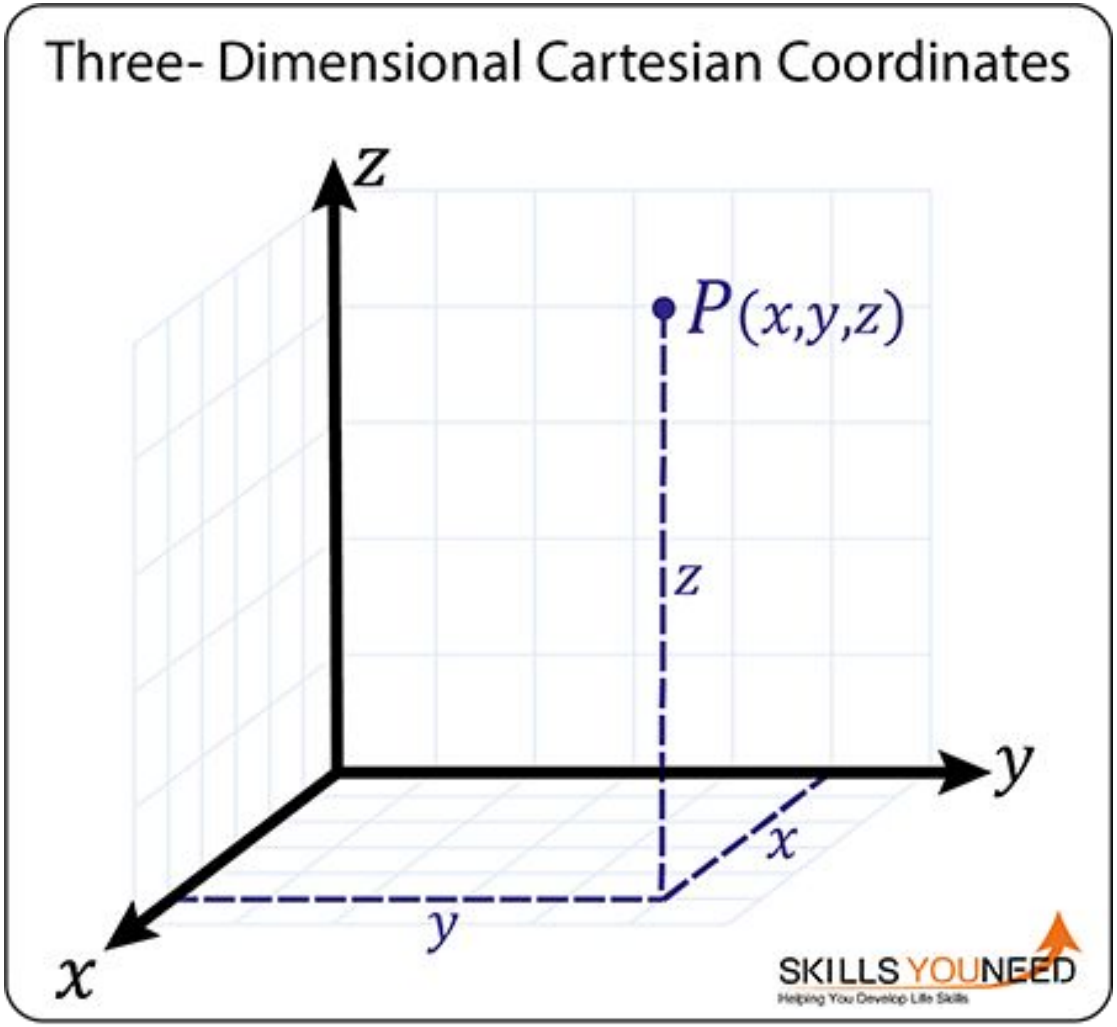
- **<u>For Coordinates C(5, 6)</u>**
- Let the new coordinates of corner C after reflection = (X_{new}, Y_{new}).
- Applying the reflection equations, we have-

$$X_{new} = X_{old} = 5$$
$$Y_{new} = -Y_{old} = -6$$

- Thus, New coordinates of corner C after reflection = (5, -6).
- **Thus, New coordinates of the triangle after reflection = A (3, -4), B(6, -4), C(5, -6).**

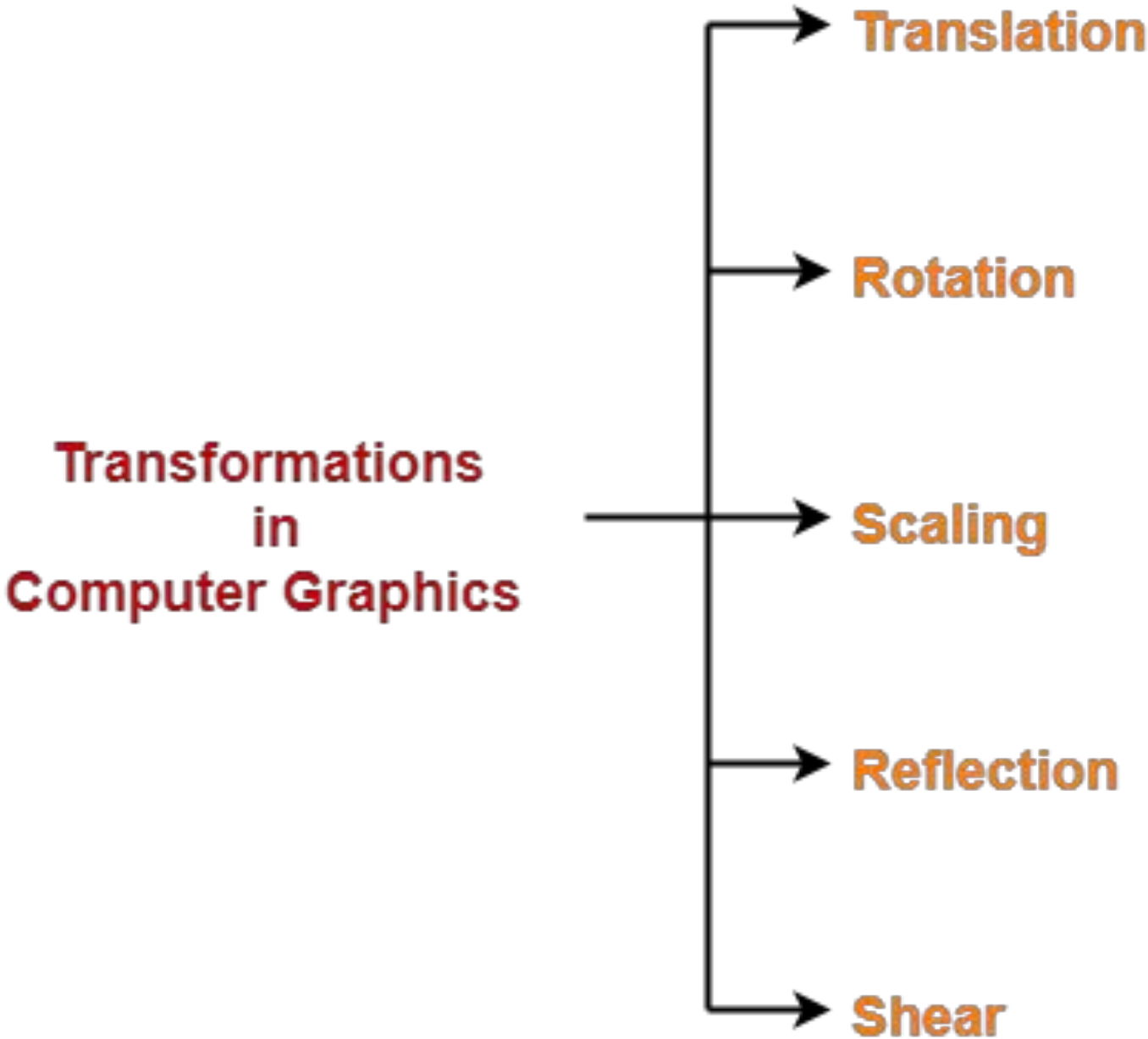

Y
C
6
Original Object
A
B
4
-X
X
Mirror
0
3
5
6
A
B
-4
Reflected Object
6
C
-Y

3D Coordinate System



3D Transformation in Computer Graphics-

- 3D Transformations take place in a three dimensional plane.
- 3D Transformations are important and a bit more complex than 2D Transformations.
- Transformations are helpful in changing the position, size, orientation, shape etc of the object.


Transformations
in
Computer Graphics
Translation
Rotation
Scaling
Reflection
Shear

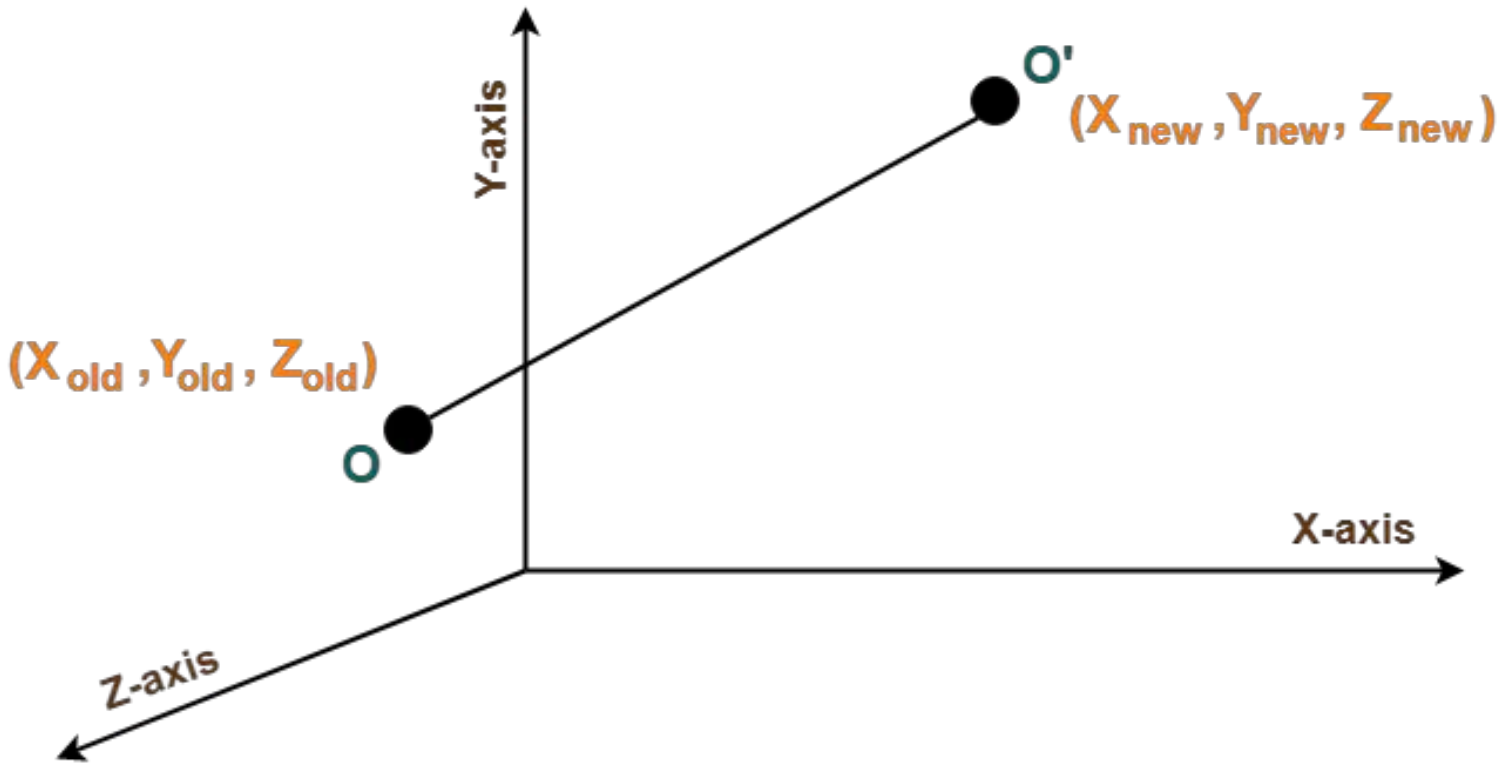
3D Translation

- In Computer graphics,3D Translation is a process of moving an object from one position to another in a three dimensional plane.
- Consider a point object O has to be moved from one position to another in a 3D plane.

 Let-

- Initial coordinates of the object O = **(Xold, Yold, Zold)**
- New coordinates of the object O after translation = **(Xnew, Ynew, Zold)**
- Translation vector or Shift vector = **(Tx, Ty, Tz)**

- Given a Translation vector (T_x, T_y, T_z)-
- T_x defines the distance the X_{old} coordinate has to be moved.
- T_y defines the distance the Y_{old} coordinate has to be moved.
- T_z defines the distance the Z_{old} coordinate has to be moved.



3D Translation in Computer Graphics

- This translation is achieved by adding the translation coordinates to the old coordinates of the object as-
- $\mathbf{X_{new} = X_{old} + T_x}$ (This denotes translation towards X axis)
- $\mathbf{Y_{new} = Y_{old} + T_y}$ (This denotes translation towards Y axis)
- $\mathbf{Z_{new} = Z_{old} + T_z}$ (This denotes translation towards Z axis)
-
- In Matrix form, the above translation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Translation Matrix

PRACTICE PROBLEM BASED ON 3D TRANSLATION IN COMPUTER GRAPHICS-

- **Problem-**
- Given a 3D object with coordinate points A(0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0). Apply the translation with the distance 1 towards X axis, 1 towards Y axis and 2 towards Z axis and obtain the new coordinates of the object.
- **Solution-**
- **Given-**
- Old coordinates of the object = A (0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0)
- Translation vector = (Tx, Ty, Tz) = (1, 1, 2)

- **<u>For Coordinates A(0, 3, 1)</u>**
- Let the new coordinates of A = (X_{new}, Y_{new}, Z_{new})
- Applying the translation equations, we have-
- Xnew = Xold + Tx = 0 + 1 = 1
- Ynew = Yold + Ty = 3 + 1 = 4
- Znew = Zold + Tz = 1 + 2 = 3
- **Thus, New coordinates of A = (1, 4, 3).**

- **<u>For Coordinates B(3, 3, 2)</u>**
- Let the new coordinates of B = (Xnew, Ynew, Znew).
- Applying the translation equations, we have-
- Xnew = Xold + Tx = 3 + 1 = 4
- Ynew = Yold + Ty = 3 + 1 = 4
- Znew = Zold + Tz = 2 + 2 = 4
- **Thus, New coordinates of B = (4, 4, 4).**

- **<u>For Coordinates C(3, 0, 0)</u>**
- Let the new coordinates of C = (X_{new}, Y_{new}, Z_{new}).
- Applying the translation equations, we have-
- $X_{new} = X_{old} + T_x = 3 + 1 = 4$
- $Y_{new} = Y_{old} + T_y = 0 + 1 = 1$
- $Z_{new} = Z_{old} + T_z = 0 + 2 = 2$
- **Thus, New coordinates of C = (4, 1, 2).**

- **<u>For Coordinates D(0, 0, 0)</u>**
- Let the new coordinates of D = (Xnew, Ynew, Znew).
- Applying the translation equations, we have-
- Xnew = Xold + Tx = 0 + 1 = 1
- Ynew = Yold + Ty = 0 + 1 = 1
- Znew = Zold + Tz = 0 + 2 = 2
- **Thus, New coordinates of D = (1, 1, 2).**

- **Thus, New coordinates of the object = A (1, 4, 3),**
  B(4, 4, 4),
  C(4, 1, 2),
  D(1, 1, 2).

3D Rotation

- In Computer graphics,3D Rotation is a process of rotating an object with respect to an angle in a three dimensional plane.
- Consider a point object O has to be rotated from one angle to another in a 3D plane.
- Let-
- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Initial angle of the object O with respect to origin = Φ
- Rotation angle = θ
- New coordinates of the object O after rotation = $(X_{new}, Y_{new}, Z_{new})$

- In 3 dimensions, there are 3 possible types of rotation-
- X-axis Rotation
- Y-axis Rotation
- Z-axis Rotation
- **<u>For X-Axis Rotation-</u>**
- This rotation is achieved by using the following rotation equations-

  - $\mathbf{X_{new} = X_{old}}$
  - $\mathbf{Y_{new} = Y_{old} \times \cos\theta - Z_{old} \times \sin\theta}$
  - $\mathbf{Z_{new} = Y_{old} \times \sin\theta + Z_{old} \times \cos\theta}$

- In Matrix form, the above rotation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Rotation Matrix

(For X-Axis Rotation)

- **<u>For Y-Axis Rotation-</u>**
- This rotation is achieved by using the following rotation equations-
  - $X_{new} = Z_{old} \times \sin\theta + X_{old} \times \cos\theta$
  - $Y_{new} = Y_{old}$
  - $Z_{new} = Y_{old} \times \cos\theta - X_{old} \times \sin\theta$
- In Matrix form, the above rotation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Rotation Matrix

(For Y-Axis Rotation)

- **<u>For Z-Axis Rotation-</u>**
- This rotation is achieved by using the following rotation equations-
  - $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$
  - $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$
  - $Z_{new} = Z_{old}$
- In Matrix form, the above rotation equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Rotation Matrix

(For Z-Axis Rotation)

- **<u>Problem-01:</u>**
- Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points.
- **<u>Solution-</u>**
- Given-
- Old coordinates = (Xold, Yold, Zold) = (1, 2, 3)
- Rotation angle = θ = 90º
- **<u>For X-Axis Rotation-</u>**
- Let the new coordinates after rotation = (X_{new}, Y_{new}, Z_{new}).

- Applying the rotation equations, we have-
- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} \times \cos\theta - Z_{old} \times \sin\theta$

  $= 2 \times \cos 90° - 3 \times \sin 90°$

  $= 2 \times 0 - 3 \times 1$

  $= -3$
- $Z_{new} = Y_{old} \times \sin\theta + Z_{old} \times \cos\theta$

  $= 2 \times \sin 90° + 3 \times \cos 90°$

  $= 2 \times 1 + 3 \times 0$

  $= 2$
- **Thus, New coordinates after rotation = (1, -3, 2).**

- **<u>For Y-Axis Rotation-</u>**
- Let the new coordinates after rotation = (X_{new}, Y_{new}, Z_{new}).
- Applying the rotation equations, we have-
- $X_{new} = Z_{old} \times \sin\theta + X_{old} \times \cos\theta$

  $= 3 \times \sin 90° + 1 \times \cos 90°$

  $= 3 \times 1 + 1 \times 0$

  $= 3$
- $Y_{new} = Y_{old} = 2$
- $Z_{new} = Y_{old} \times \cos\theta - X_{old} \times \sin\theta$

  $= 2 \times \cos 90° - 1 \times \sin 90°$

  $= 2 \times 0 - 1 \times 1$

  $= -1$
- **Thus, New coordinates after rotation = (3, 2, -1).**

- **<u>For Z-Axis Rotation-</u>**
- Let the new coordinates after rotation = (X_{new}, Y_{new}, Z_{new}).
- Applying the rotation equations, we have-
- $X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$

  $= 1 \times \cos 90° - 2 \times \sin 90°$

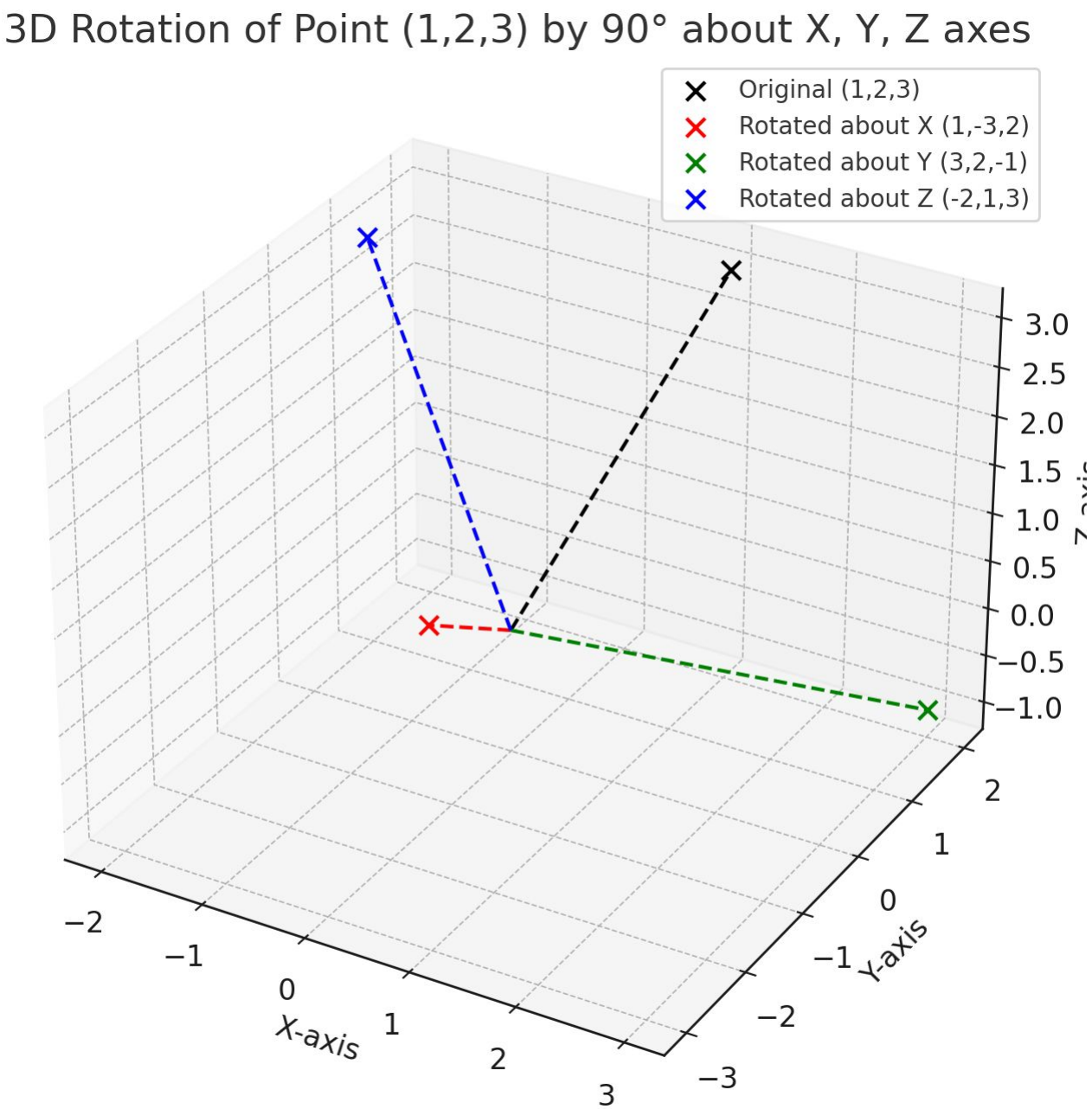
  $= 1 \times 0 - 2 \times 1$

  $= -2$
- $Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$

  $= 1 \times \sin 90° + 2 \times \cos 90°$

  $= 1 \times 1 + 2 \times 0$

  $= 1$
- $Z_{new} = Z_{old} = 3$
- **Thus, New coordinates after rotation = (-2, 1, 3).**


3D Rotation of Point (1,2,3) by 90° about X, Y, Z axes
Original (1,2,3)
Rotated about X (1,-3,2)
Rotated about Y (3,2,-1)
Rotated about Z (-2,1,3)
X-axis
Y-axis
Z-axis
−2
−1
0
1
2
3
−3
−2
−1
0
1
2
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

3D Scaling

- In computer graphics, scaling is a process of modifying or altering the size of objects.
- Scaling may be used to increase or reduce the size of object.
- Scaling subjects the coordinate points of the original object to change.
- Scaling factor determines whether the object size is to be increased or reduced.
- If scaling factor > 1, then the object size is increased.
- If scaling factor < 1, then the object size is reduced.

- Consider a point object O has to be scaled in a 3D plane.
- Let-
- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Scaling factor for X-axis = S_x
- Scaling factor for Y-axis = S_y
- Scaling factor for Z-axis = S_z
- New coordinates of the object O after scaling = $(X_{new}, Y_{new}, Z_{new})$

- This scaling is achieved by using the following scaling equations-

- $\mathbf{X_{new} = X_{old} \times S_x}$
- $\mathbf{Y_{new} = Y_{old} \times S_y}$
- $\mathbf{Z_{new} = Z_{old} \times S_z}$

- In Matrix form, the above scaling equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Scaling Matrix

- **<u>Problem-01:</u>**
- Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0). Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object.

- **<u>Solution-</u>**
- Given-
- Old coordinates of the object = A (0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0)
- Scaling factor along X axis = 2
- Scaling factor along Y axis = 3
- Scaling factor along Z axis = 3
- **<u>For Coordinates A(0, 3, 3)</u>**
- Let the new coordinates of A after scaling = (X_{new}, Y_{new}, Z_{new}).

- Applying the scaling equations, we have-
- $X_{new} = X_{old} \times S_x = 0 \times 2 = 0$
- $Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$
- $Z_{new} = Z_{old} \times S_z = 3 \times 3 = 9$
- **Thus, New coordinates of corner A after scaling = (0, 9, 9).**

- **<u>For Coordinates B(3, 3, 6)</u>**
- Let the new coordinates of B after scaling = $(X_{new}, Y_{new}, Z_{new})$.
- Applying the scaling equations, we have-
- $X_{new} = X_{old} \times S_x = 3 \times 2 = 6$
- $Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$
- $Z_{new} = Z_{old} \times S_z = 6 \times 3 = 18$

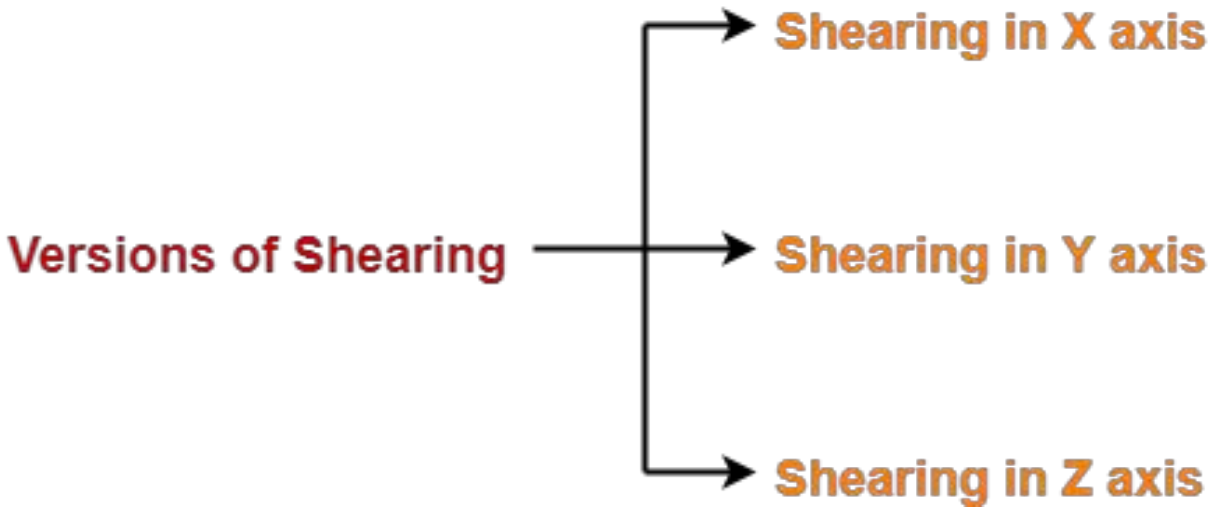
- **Thus, New coordinates of corner B after scaling = (6, 9, 18).**

- **<u>For Coordinates C(3, 0, 1)</u>**
- Let the new coordinates of C after scaling = (X_{new}, Y_{new}, Z_{new}).
- Applying the scaling equations, we have-
  - $X_{new} = X_{old} \times S_x = 3 \times 2 = 6$
  - $Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$
  - $Z_{new} = Z_{old} \times S_z = 1 \times 3 = 3$
- Thus, New coordinates of corner C after scaling = (6, 0, 3).

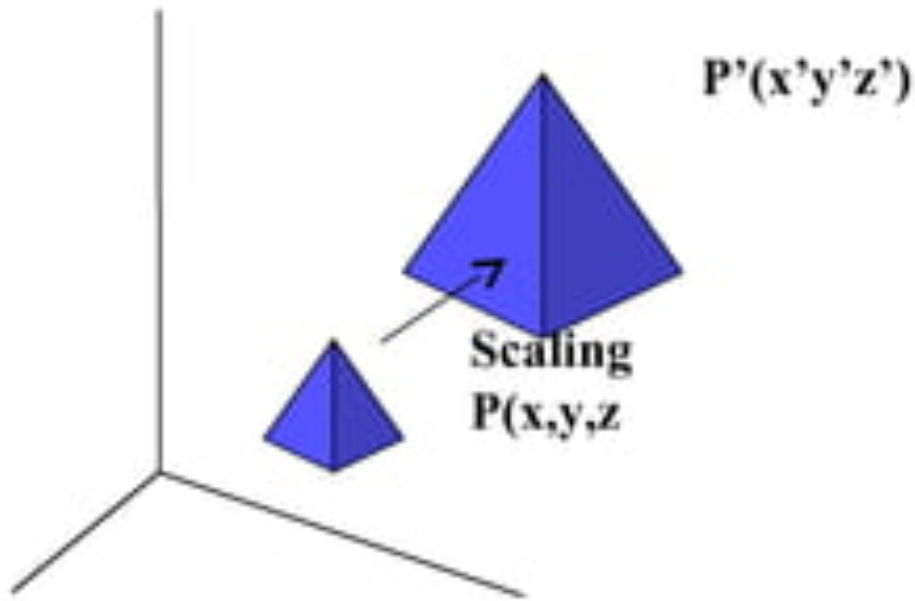
- **<u>For Coordinates D(0, 0, 0)</u>**
- Let the new coordinates of D after scaling = (X_{new}, Y_{new}, Z_{new}).
- Applying the scaling equations, we have-
- $X_{new} = X_{old} \times S_x = 0 \times 2 = 0$
- $Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$
- $Z_{new} = Z_{old} \times S_z = 0 \times 3 = 0$
- Thus, New coordinates of corner D after scaling = (0, 0, 0).

3D Shearing

- In Computer graphics,3D Shearing is an ideal technique to change the shape of an existing object in a three dimensional plane.
- In a three dimensional plane, the object size can be changed along X direction, Y direction as well as Z direction.
- So, **there are three versions of shearing-**



Sx, Sy and Sz are scaling factors along x, y and z directions..



- Consider a point object O has to be sheared in a 3D plane.
- Let-
- Initial coordinates of the object O = $\mathbf{(X_{old}, Y_{old}, Z_{old})}$
- Shearing parameter towards X direction = $\mathbf{Sh_x}$
- Shearing parameter towards Y direction = $\mathbf{Sh_y}$
- Shearing parameter towards Z direction = $\mathbf{Sh_z}$
- New coordinates of the object O after shearing = $\mathbf{(X_{new}, Y_{new}, Z_{new})}$

- **<u>1) Shearing in X Axis-</u>**
- Shearing in X axis is achieved by using the following shearing equations-
  - $X_{new} = X_{old}$
  - $Y_{new} = Y_{old} + Sh_y \times X_{old}$
  - $Z_{new} = Z_{old} + Sh_z \times X_{old}$
- In Matrix form, the above shearing equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ Sh_y & 1 & 0 & 0 \\ Sh_z & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In X axis)

- **<u>Shearing in Y Axis-</u>**
- Shearing in Y axis is achieved by using the following shearing equations-

  - $X_{new} = X_{old} + Sh_x \times Y_{old}$
  - $Y_{new} = Y_{old}$
  - $Z_{new} = Z_{old} + Sh_z \times Y_{old}$
- In Matrix form, the above shearing equations may be represented as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Sh_x & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & Sh_z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In Y axis)

- **<u>Shearing in Z Axis-</u>**
- Shearing in Z axis is achieved by using the following shearing equations-

  - $X_{new} = X_{old} + Sh_x \times Z_{old}$
  - $Y_{new} = Y_{old} + Sh_y \times Z_{old}$
  - $Z_{new} = Z_{old}$

$$\begin{bmatrix} X_{new} \\ Y_{new} \\ Z_{new} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & Sh_x & 0 \\ 0 & 1 & Sh_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \\ Z_{old} \\ 1 \end{bmatrix}$$

3D Shearing Matrix

(In Z axis)

- **<u>Problem-01:</u>**
- Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.
- **Solution-**
- **Given-**
- Old corner coordinates of the triangle = A (0, 0, 0), B(1, 1, 2), C(1, 1, 3)
- Shearing parameter towards X direction (Shx) = 2
- Shearing parameter towards Y direction (Shy) = 2
- Shearing parameter towards Y direction (Shz) = 3

- **<u>Shearing in X Axis-</u>**
- **<u>For Coordinates A(0, 0, 0)</u>**
- Let the new coordinates of corner A after shearing = $(X_{new}, Y_{new}, Z_{new})$.
- Applying the shearing equations, we have-
  - $X_{new} = X_{old} = 0$
  - $Y_{new} = Y_{old} + Sh_y \times X_{old} = 0 + 2 \times 0 = 0$
  - $Z_{new} = Z_{old} + Sh_z \times X_{old} = 0 + 3 \times 0 = 0$
- Thus, New coordinates of corner A after shearing = (0, 0, 0).
- **<u>For Coordinates B(1, 1, 2)</u>**
- Let the new coordinates of corner B after shearing = $(X_{new}, Y_{new}, Z_{new})$.

- Applying the shearing equations, we have-
- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} + Sh_y \times X_{old} = 1 + 2 \times 1 = 3$
- $Z_{new} = Z_{old} + Sh_z \times X_{old} = 2 + 3 \times 1 = 5$
- Thus, New coordinates of corner B after shearing = (1, 3, 5).
- **<u>For Coordinates C(1, 1, 3)</u>**
- Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}, Z_{new}).
- Applying the shearing equations, we have-
- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} + Sh_y \times X_{old} = 1 + 2 \times 1 = 3$
- $Z_{new} = Z_{old} + Sh_z \times X_{old} = 3 + 3 \times 1 = 6$
- Thus, New coordinates of corner C after shearing = (1, 3, 6).
- Thus, New coordinates of the triangle after shearing in X axis = A (0, 0, 0), B(1, 3, 5), C(1, 3, 6).

- **<u>Shearing in Y Axis-</u>**
- **<u>For Coordinates A(0, 0, 0)</u>**
- Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}, Z_{new}).
- Applying the shearing equations, we have-
- $X_{new} = X_{old} + Sh_x \times Y_{old} = 0 + 2 \times 0 = 0$
- $Y_{new} = Y_{old} = 0$
- $Z_{new} = Z_{old} + Sh_z \times Y_{old} = 0 + 3 \times 0 = 0$
- Thus, New coordinates of corner A after shearing = (0, 0, 0).
- **<u>For Coordinates B(1, 1, 2)</u>**
- Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}, Z_{new}).
- Applying the shearing equations, we have-

- $X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 1 = 3$
- $Y_{new} = Y_{old} = 1$
- $Z_{new} = Z_{old} + Sh_z \times Y_{old} = 2 + 3 \times 1 = 5$
- Thus, New coordinates of corner B after shearing = (3, 1, 5).

- **<u>For Coordinates C(1, 1, 3)</u>**
- Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}, Z_{new}).

- Applying the shearing equations, we have-
- $X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 1 = 3$
- $Y_{new} = Y_{old} = 1$
- $Z_{new} = Z_{old} + Sh_z \times Y_{old} = 3 + 3 \times 1 = 6$

- Thus, New coordinates of corner C after shearing = (3, 1, 6).
- Thus, New coordinates of the triangle after shearing in Y axis = A (0, 0, 0), B(3, 1, 5), C(3, 1, 6).

- **Shearing in Z Axis-**
- **For Coordinates A(0, 0, 0)**
- Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}, Z_{new}).
- Applying the shearing equations, we have-
- $X_{new} = X_{old} + Sh_x \times Z_{old} = 0 + 2 \times 0 = 0$
- $Y_{new} = Y_{old} + Sh_y \times Z_{old} = 0 + 2 \times 0 = 0$
- $Z_{new} = Z_{old} = 0$
- Thus, New coordinates of corner A after shearing = (0, 0, 0).

- **For Coordinates B(1, 1, 2)**
- Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}, Z_{new}).
- Applying the shearing equations, we have-
- $X_{new} = X_{old} + Sh_x \times Z_{old} = 1 + 2 \times 2 = 5$
- $Y_{new} = Y_{old} + Sh_y \times Z_{old} = 1 + 2 \times 2 = 5$
- $Z_{new} = Z_{old} = 2$
- Thus, New coordinates of corner B after shearing = (5, 5, 2).

- **<u>For Coordinates C(1, 1, 3)</u>**
- Let the new coordinates of corner C after shearing = $(X_{new}, Y_{new}, Z_{new})$.
- Applying the shearing equations, we have-
- $X_{new} = X_{old} + Sh_x \times Z_{old} = 1 + 2 \times 3 = 7$
- $Y_{new} = Y_{old} + Sh_y \times Z_{old} = 1 + 2 \times 3 = 7$
- $Z_{new} = Z_{old} = 3$
- Thus, New coordinates of corner C after shearing = (7, 7, 3).
- **Thus, New coordinates of the triangle after shearing in Z axis = A (0, 0, 0), B(5, 5, 2), C(7, 7, 3).**

Rotation about Arbitrary Axis

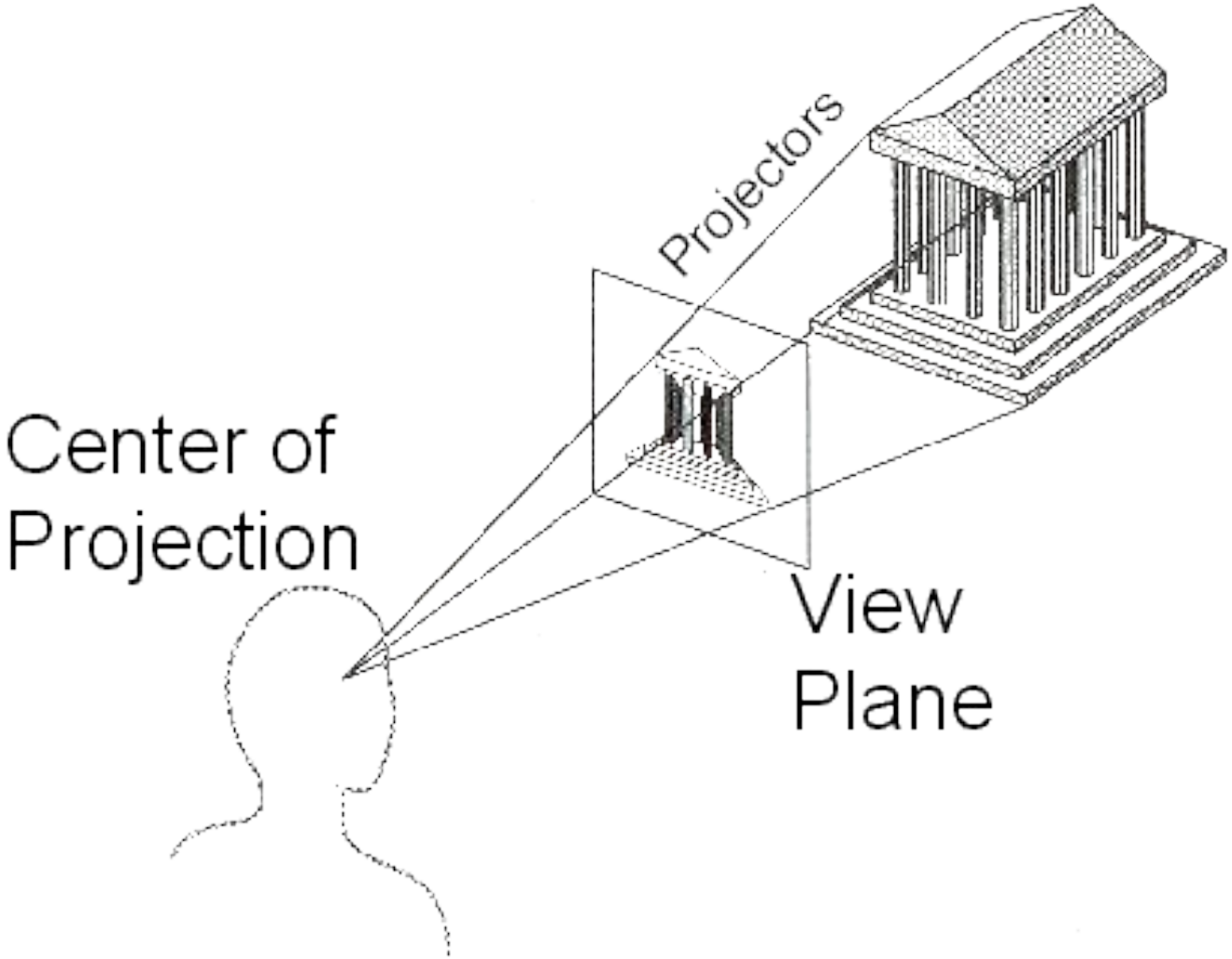
- When the object is rotated about an axis that is not parallel to any one of co-ordinate axis, i.e., x, y, z. Then additional transformations are required. First of all, alignment is needed, and then the object is being back to the original position.
- **Following steps are required**
- Translate the object to the origin
- Rotate object so that axis of object coincide with any of coordinate axis.
- Perform rotation about co-ordinate axis with whom coinciding is done.
- Apply inverse rotation to bring rotation back to the original position.

Rotation about Arbitrary Axis

Step-by-Step Procedure

- **Translate the axis to the origin:** Shift the object so that a point on the arbitrary axis moves to the origin (0,0,0) of the coordinate system.
- **Align the axis to a standard axis:**
  - Rotate space about the x-axis so the rotation axis lies in the xz-plane.
  - Rotate space about the y-axis so the rotation axis now coincides with the z-axis.
- **Perform the desired rotation:** Execute the rotation by the specified angle (theta) about the z-axis.
- **Apply inverse transformations:** Perform the inverse of the previous steps in reverse order to return the axis and object to their original position. This includes the inverse of the y-axis rotation, then the inverse of the x-axis rotation, and finally the inverse of the translation.

Projections



Projections

- **Projection is a technique to transform a 3D object into a 2D plane**.
- i.e plane {(x,y,z)---->(x,y)}
- There are two main types of projection:

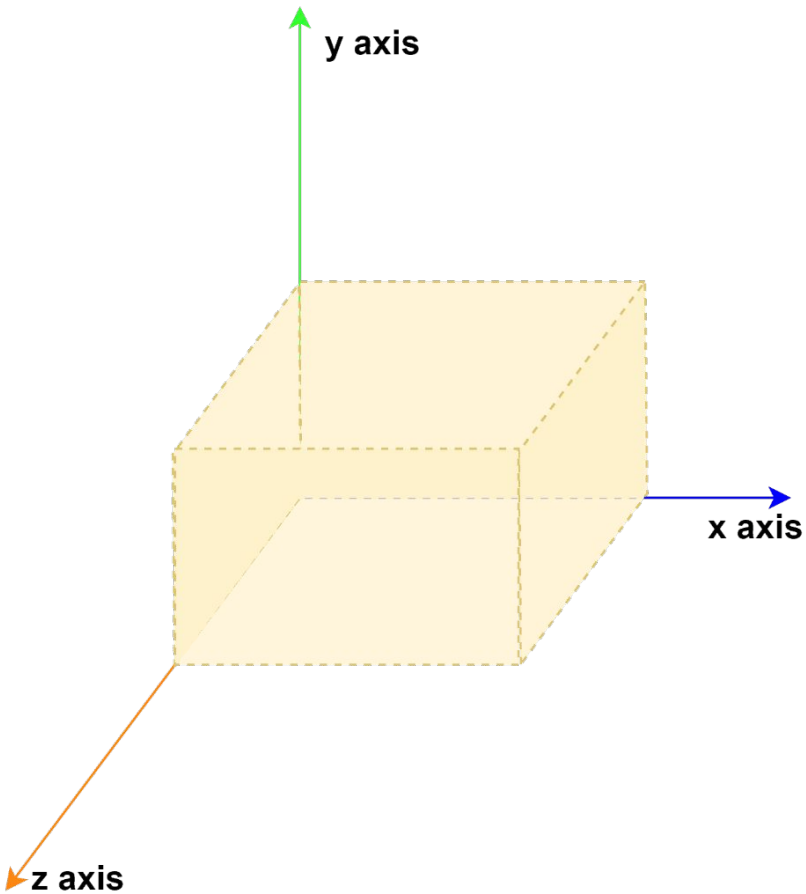
1. **Parallel**

   1.1) Orthographic: isometric, diametric, trimetric

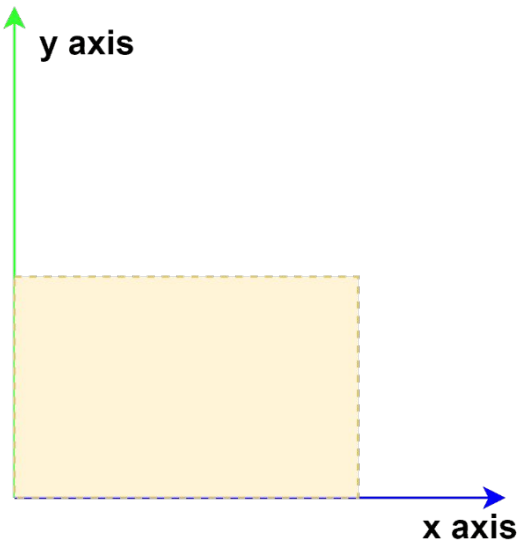
   1.2) Oblique: Cavalier, Cabinet

2. **Perspective**

   2.1) Vanishing Points – 1 point, 2 point and 3 point



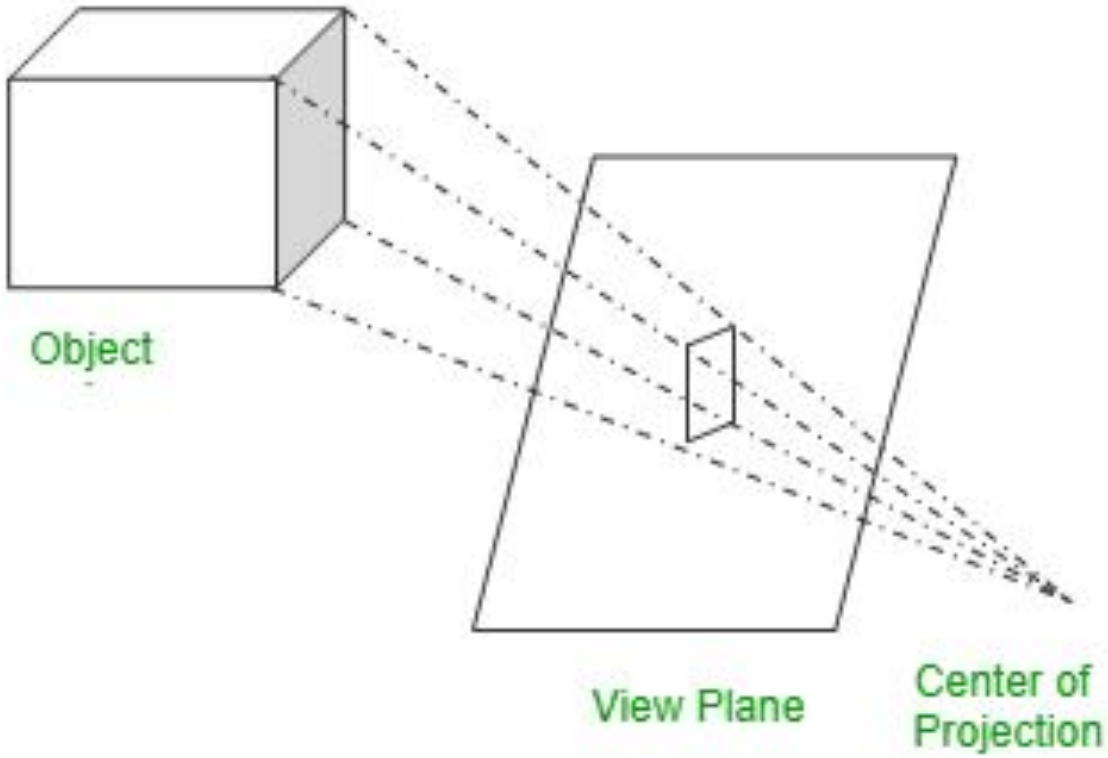
3D Figure

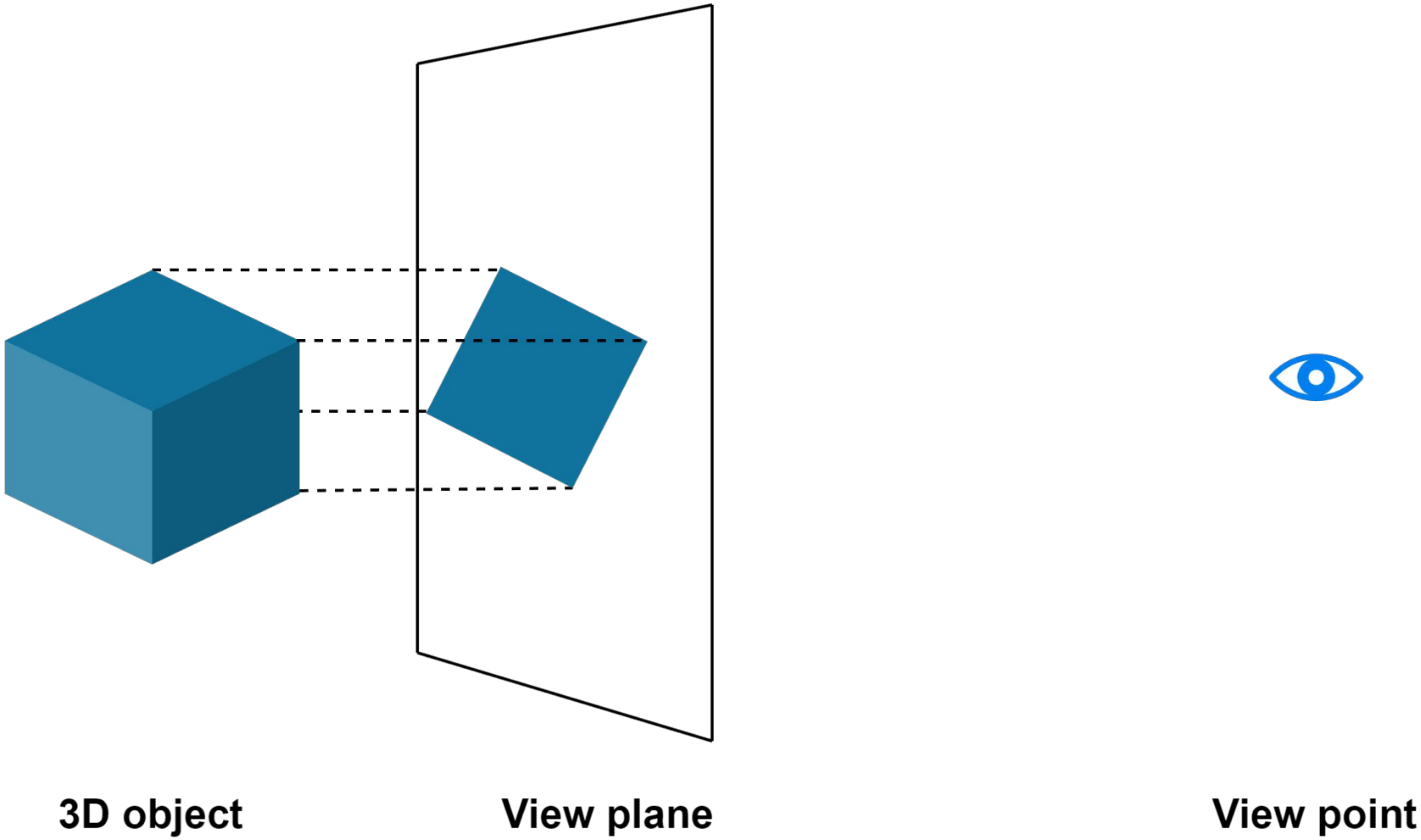


2D Figure

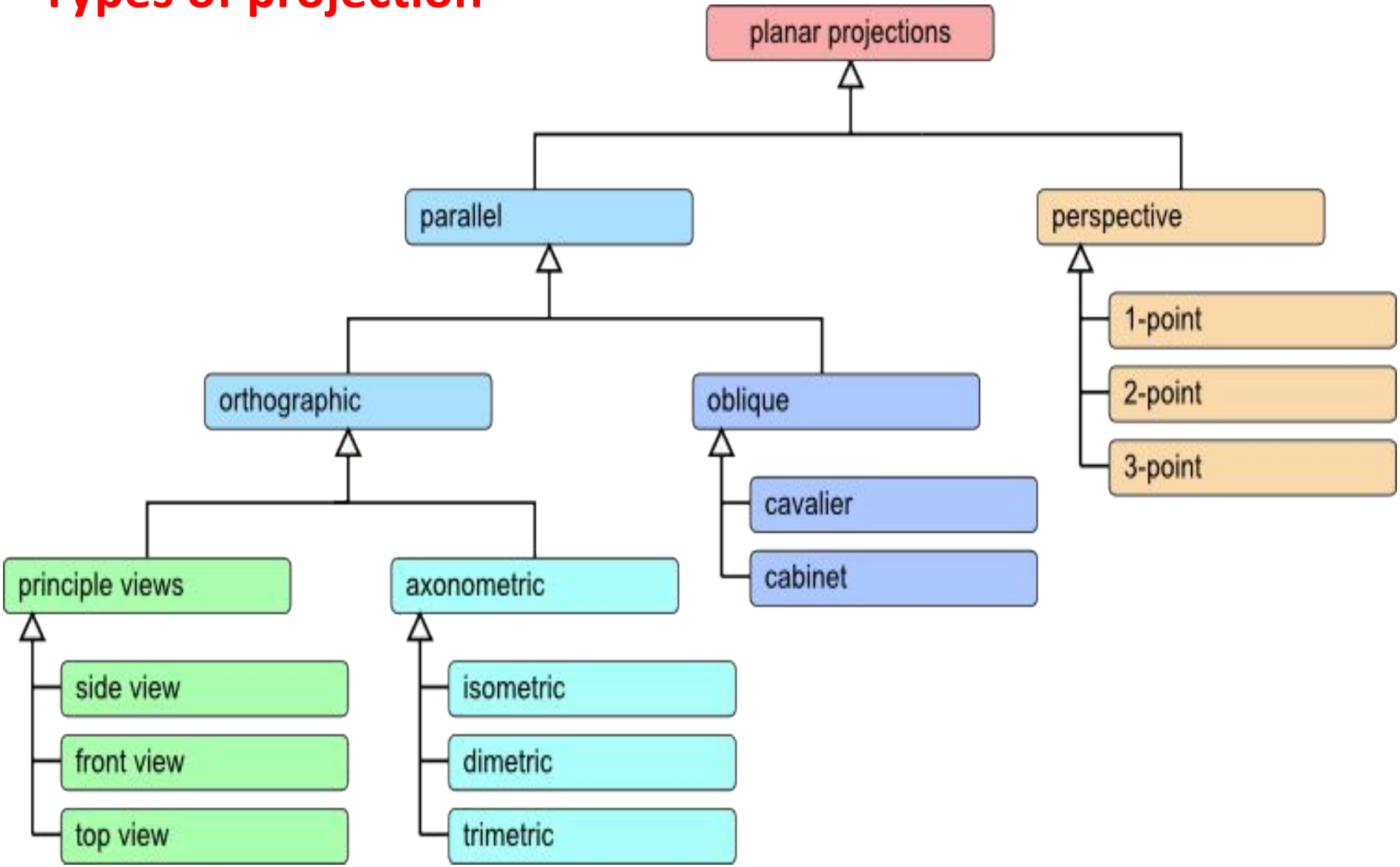
- **Projection Plane or View plane:**

"When geometric objects are formed by the intersection of lines with a plane, the plane is called the projection plane and the lines are called projections".




3D object
View plane
View point

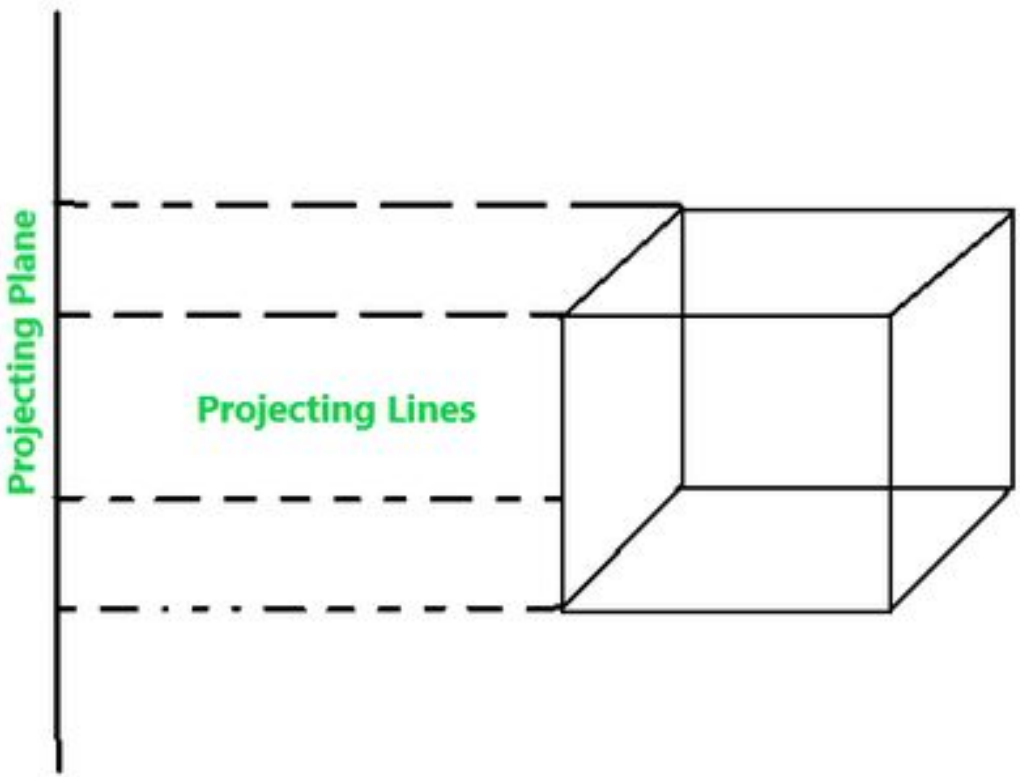
Types of projection



1) Parallel Projection

- Parallel Projection use to display picture in its true shape and size.
- The parallel projection is formed by extending parallel lines from each vertex on the object until they intersect the plane of the screen.
- The point of intersection is the projection of vertex.
- Z axis is discarded.

1) Parallel Projection

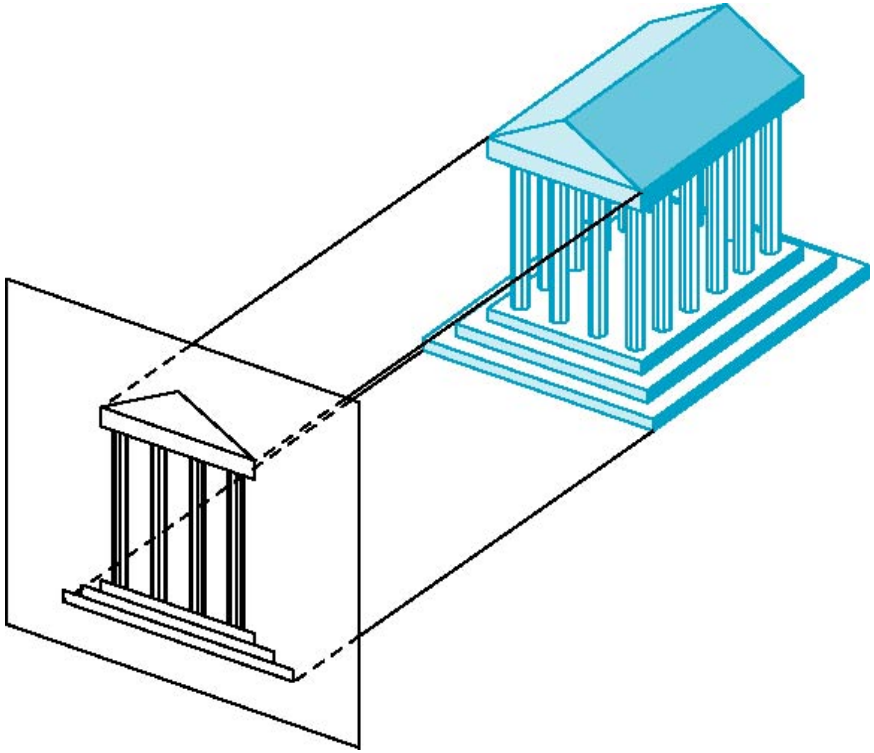


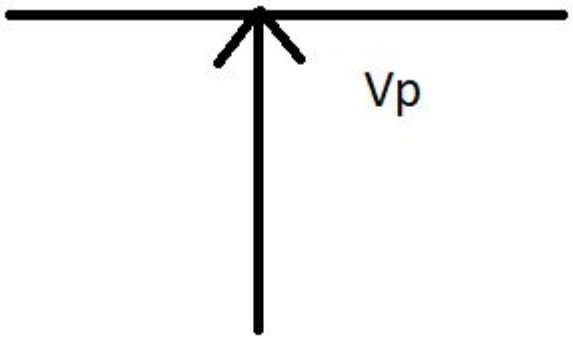
Parallel projection is further divided into two categories :

a) Orthographic Projection
b) Oblique Projection

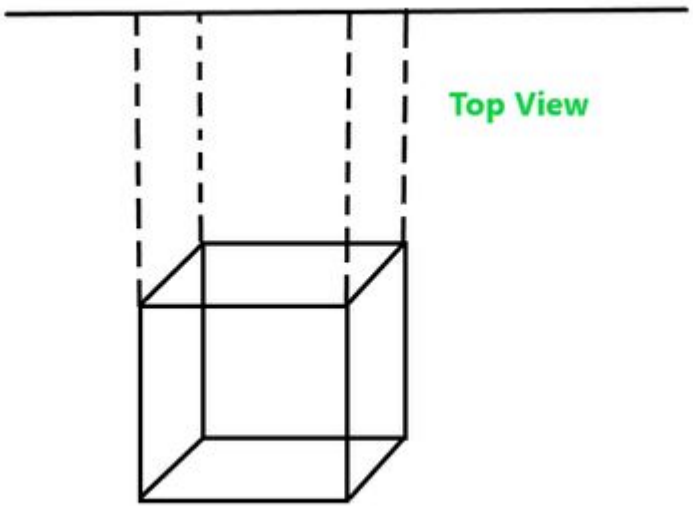
1.1) Orthographic parallel projection

- Projectors are **orthogonal** to projection surface
- In orthographic lines are parallel to each other making an angle 90 with view plane.



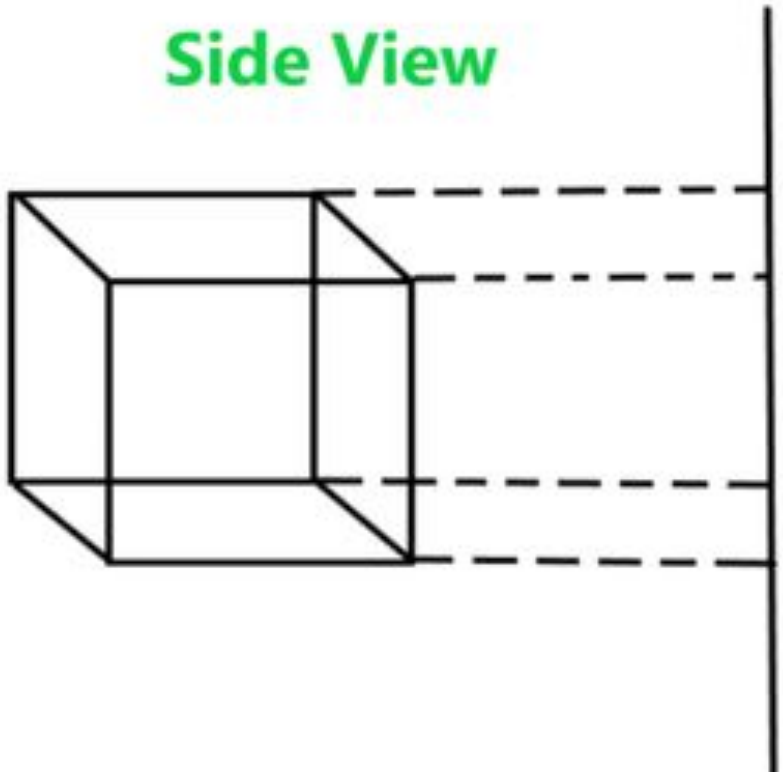


- **Orthographic Projection is of two categories :**
- **(a) Multiview Projection :** It is further divided into three categories –
- **(a.1) Top-View :** In this projection, the rays that emerge from the top of the polygon surface are observed.

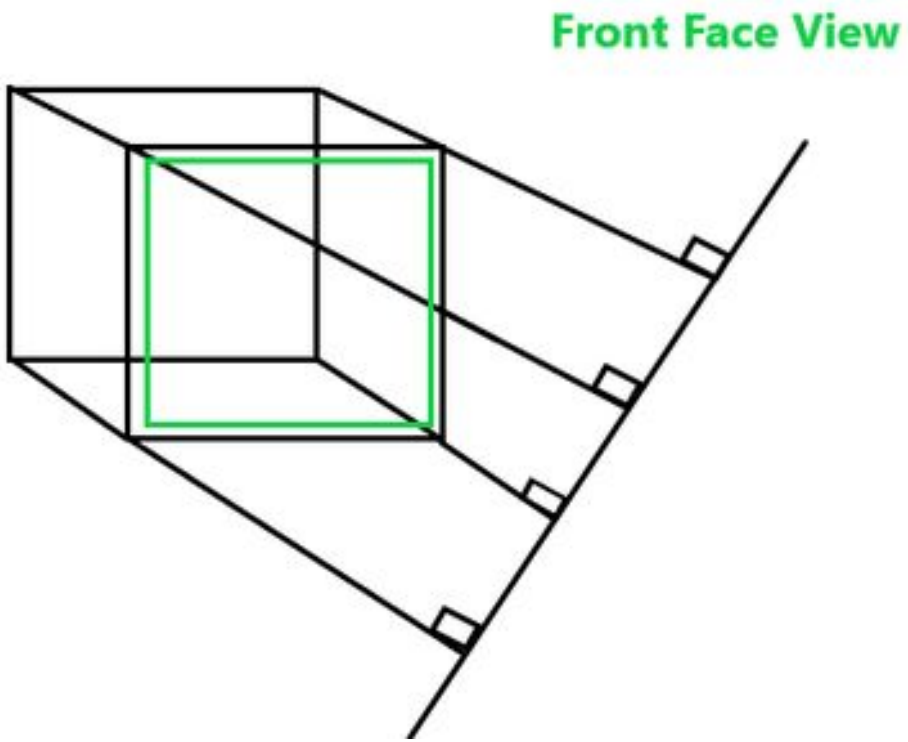


a.2) Side-View :

It is another type of projection orthographic projection where the side view of the polygon surface is
observed.



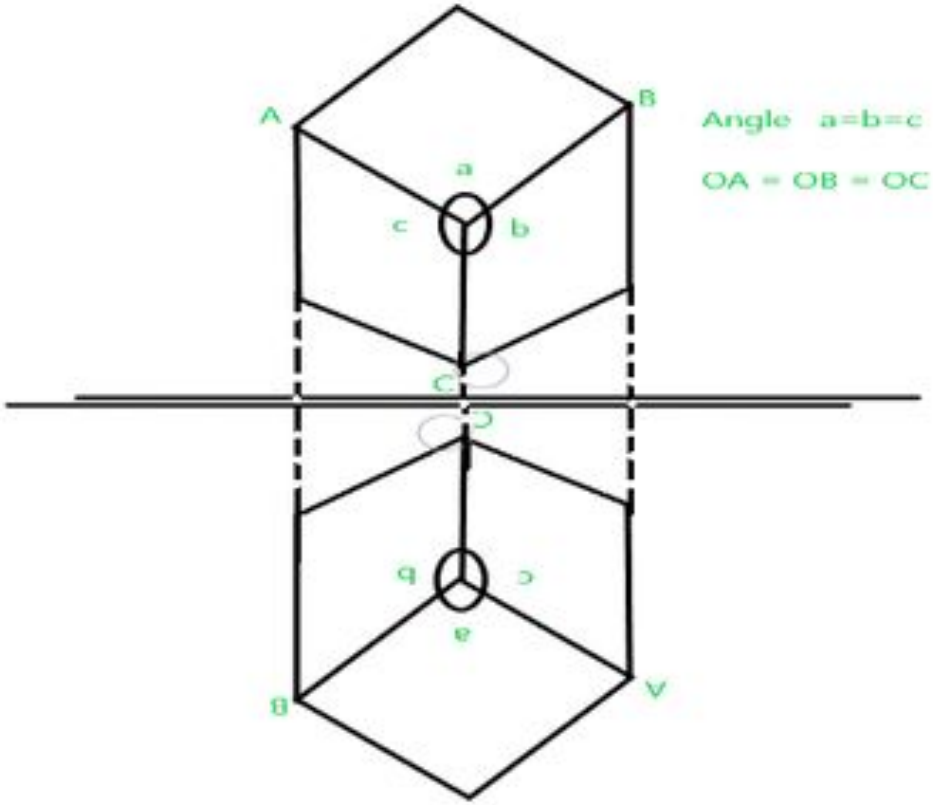
a.3) <u>Front-view</u> : In this orthographic projection front face view of the object is observed.



- **b) <u>Axonometric</u> :**
- Axonometric projection is an orthographic projection, where the **projection lines are perpendicular to the plane of projection, and the object is rotated around one or more of its axes to show multiple sides.**
- It is further divided into three categories :
- **B.1) Isometric**
- **B.2) Dimetric**
- **B.3) Trimetric**

(b.1) Isometric Projection :

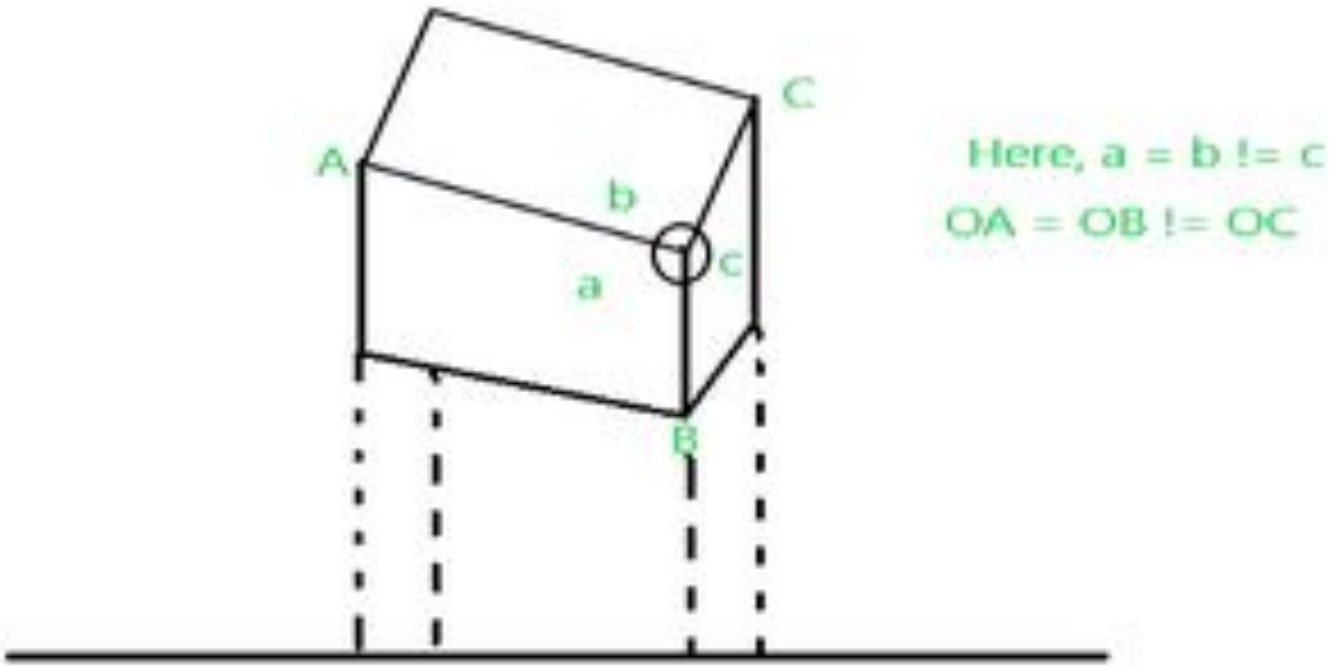
- It is a method for visually representing three-dimensional objects in two-dimensional display in technical and engineering drawings.
- Here in this projection, **the three coordinate axes appear equally foreshortened** and the angle between any two of them is 120 degrees.



Angle a=b=c

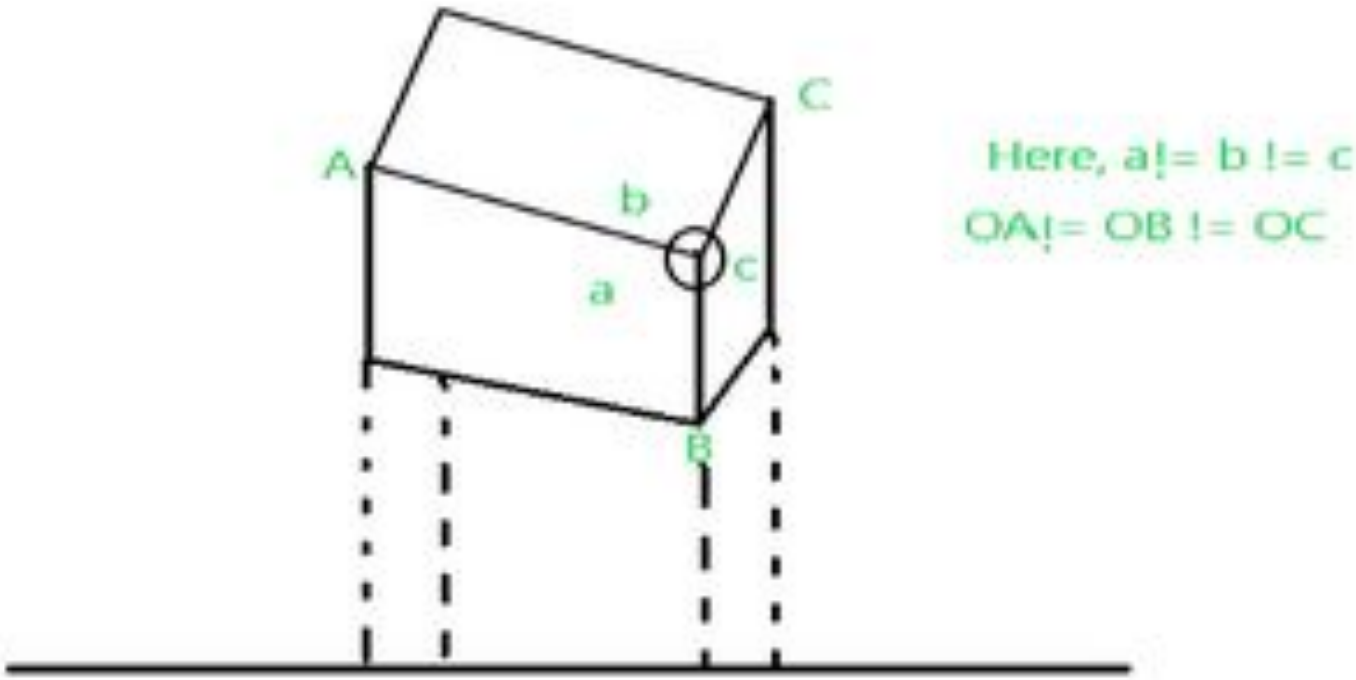
OA = OB = OC

(b.2) Dimetric Projection : It is a kind of orthographic projection where the visualized object appears to have **only two adjacent sides and angles are equal.**



Dimetric Projection

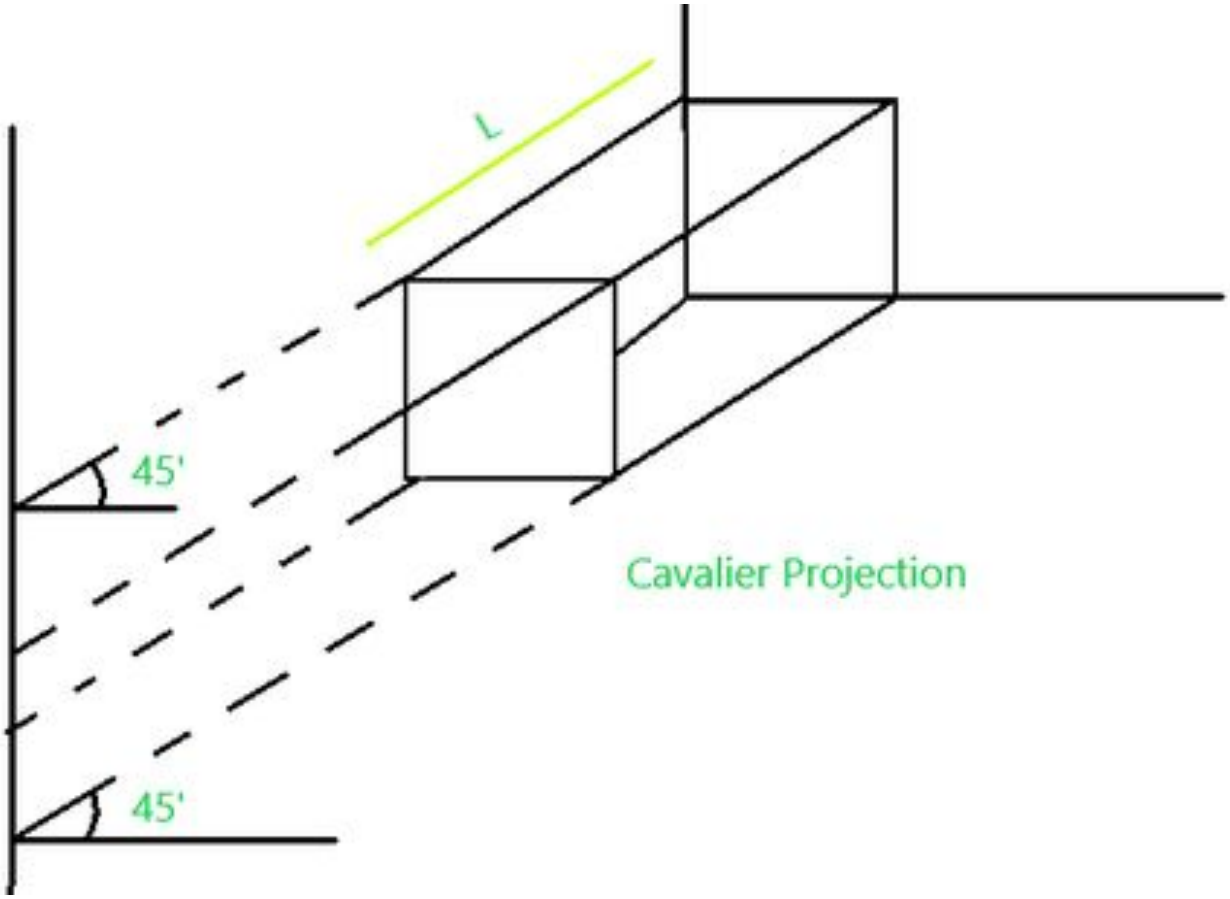
- **(b.3) Trimetric Projection :** It is a kind of orthographic projection where the visualized object appears to have **all the adjacent sides and angles unequal**.



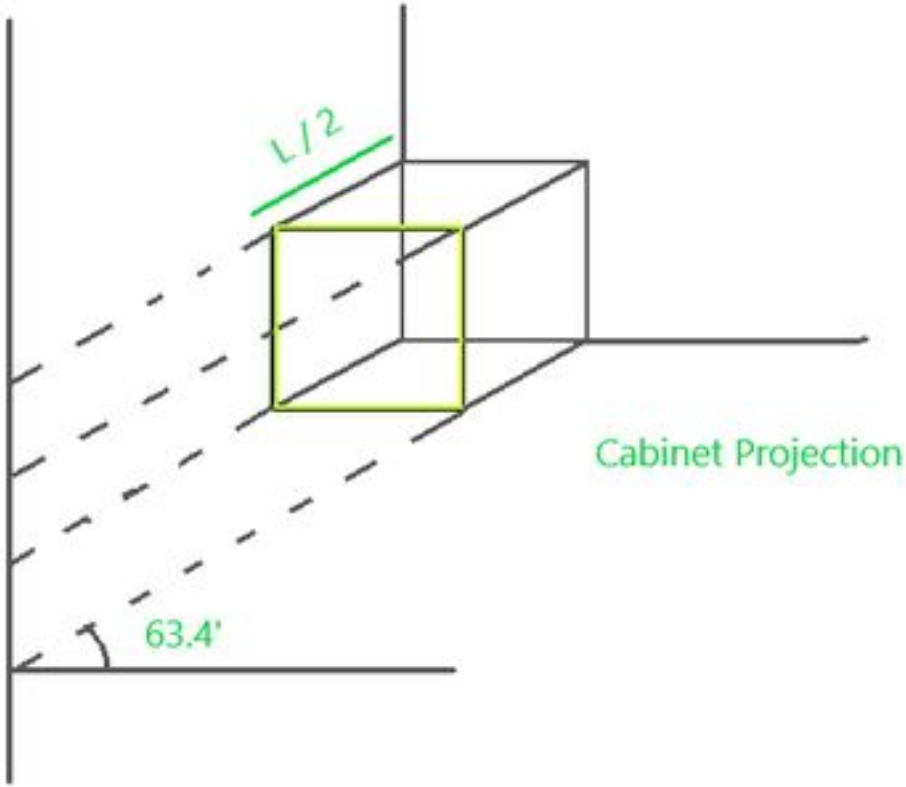
Trimetric Projection

(b) Oblique Projection

- **Oblique Projection :** It is a kind of **parallel projection** where projecting rays emerges parallelly from the surface of the polygon and incident at an angle other than 90 degrees on the plane.
- It is of two kinds :
- **(b).1. Cavalier Projection :** It is a kind of oblique projection where the projecting lines emerge parallelly from the object surface and incident at **45‘ rather than 90’** at the projecting plane. In this projection, the length of the reading axis is larger than the cabinet projection.

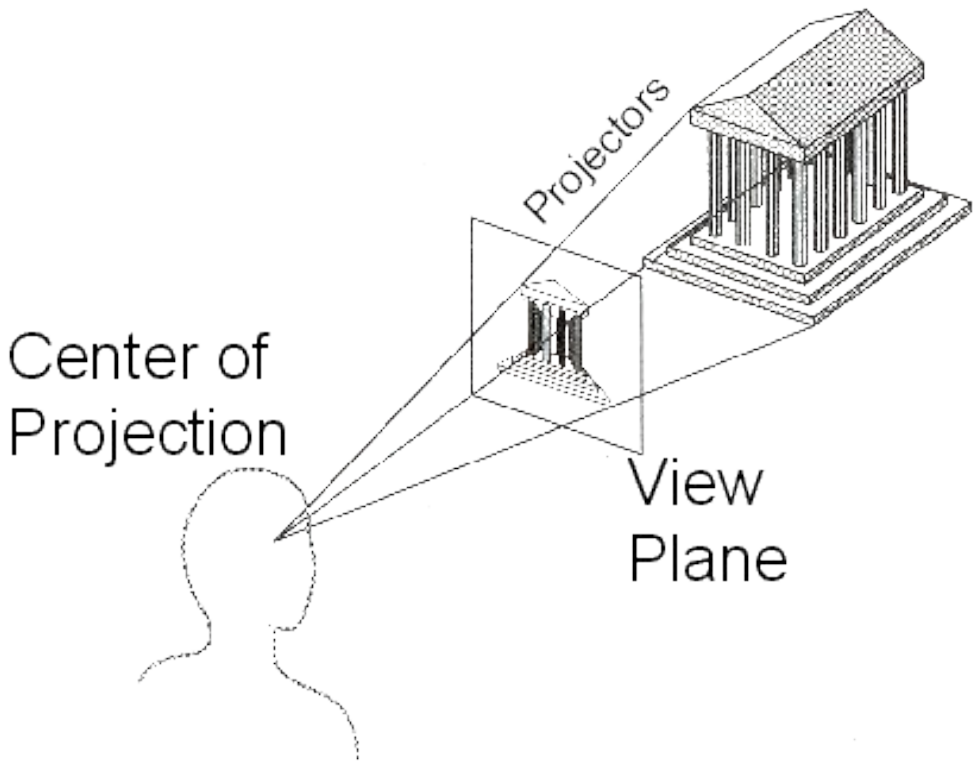

L
45'
45'
Cavalier Projection

- **(b). 2. Cabinet Projection** : It is similar to that cavalier projection but here the length of reading axes just half than the cavalier projection and the incident angle at the projecting plane is **63.4’ rather 45’.**

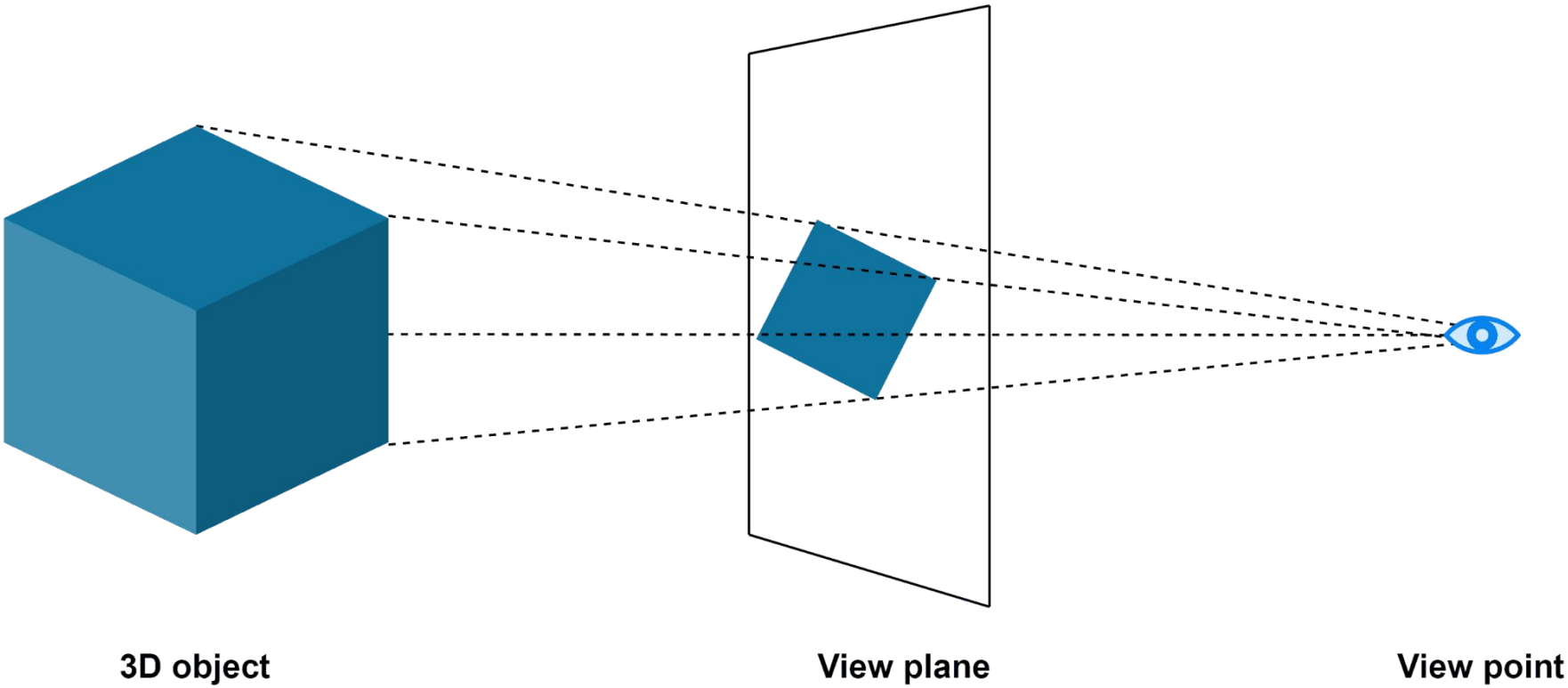


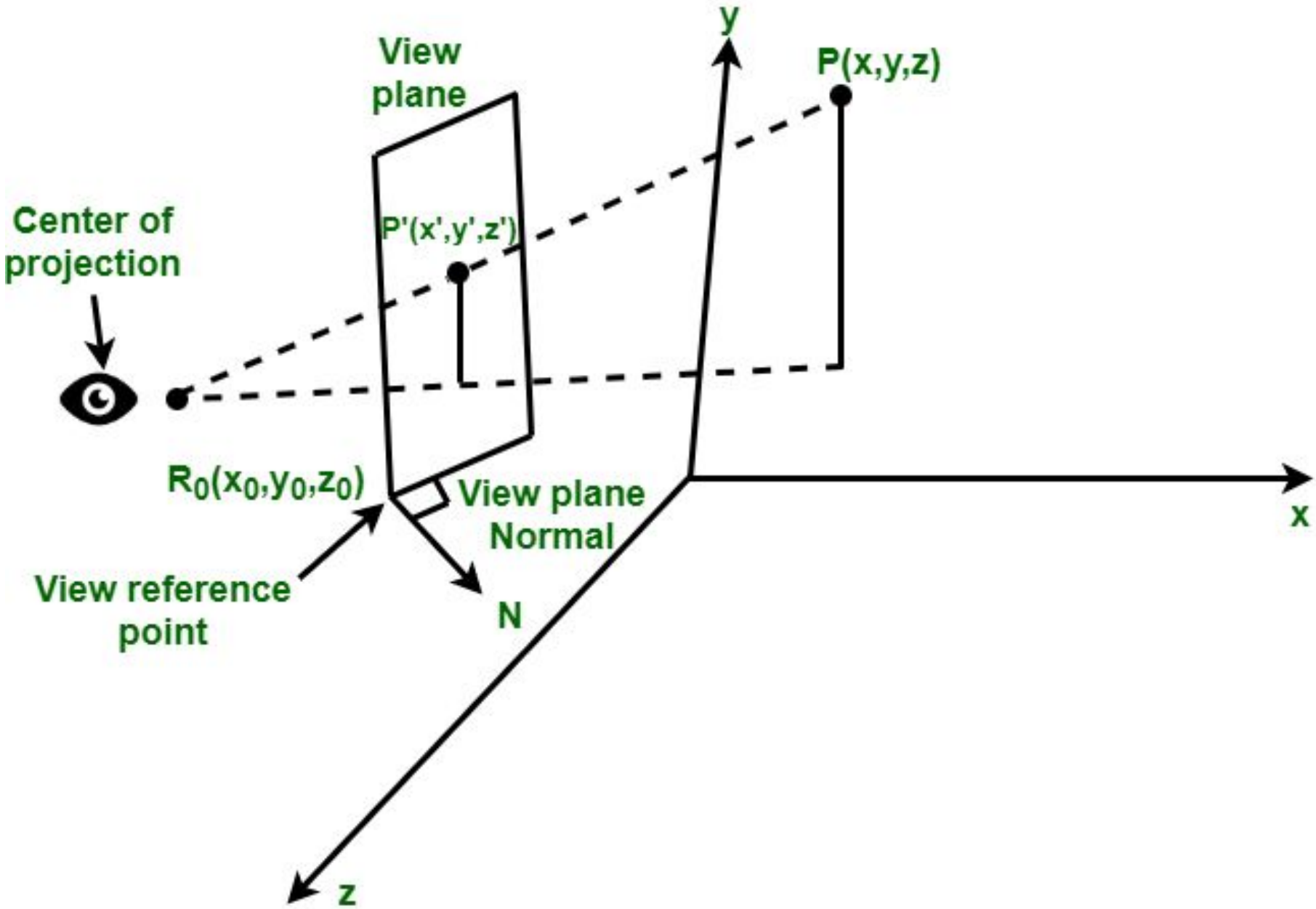
Perspective Projection

(Vanishing Points – 1 point, 2 point and 3 point)

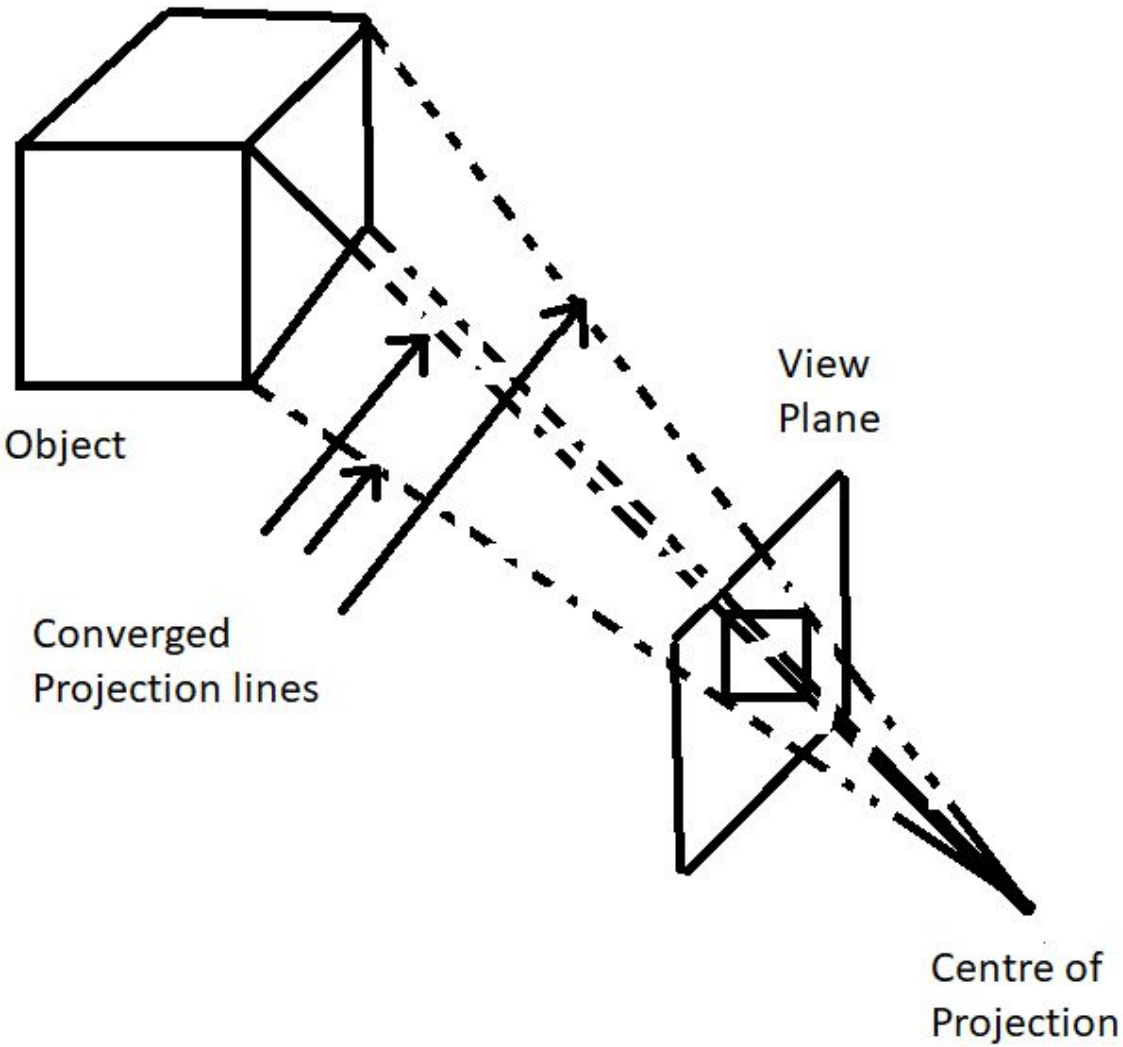


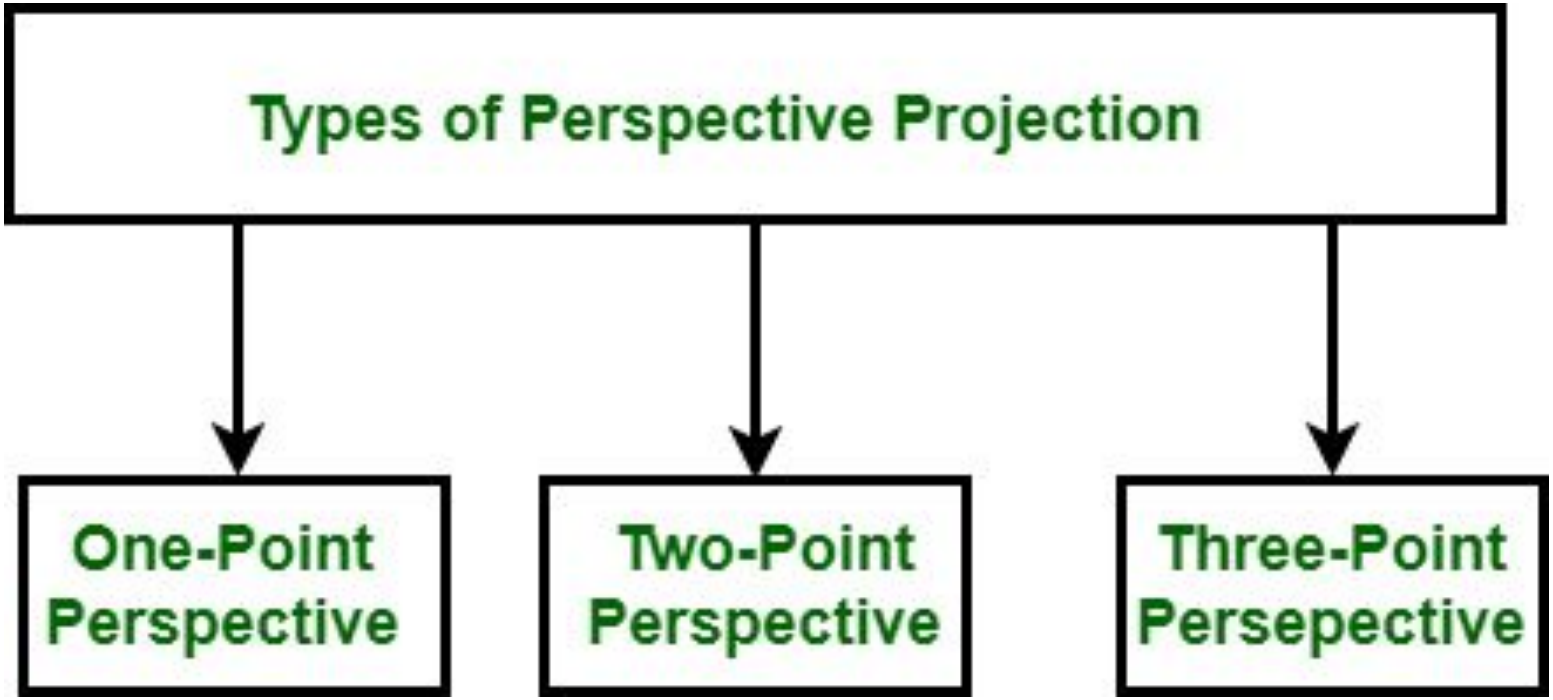
- ## 2) Perspective Projection
- In Perspective Projection the **center of projection** is at finite distance from **projection plane**.
- This projection produces realistic views but does not preserve relative proportions of an object dimensions.
- Projections of distant object **are smaller** than projections of objects of same size that are closer to projection plane.
- The perspective projection can be easily described by the following figure


3D object
View plane
View point


View
plane
y
P(x,y,z)
Center of
projection
P'(x',y',z')
$R_0(x_0,y_0,z_0)$
View plane
Normal
x
View reference
point
N
z

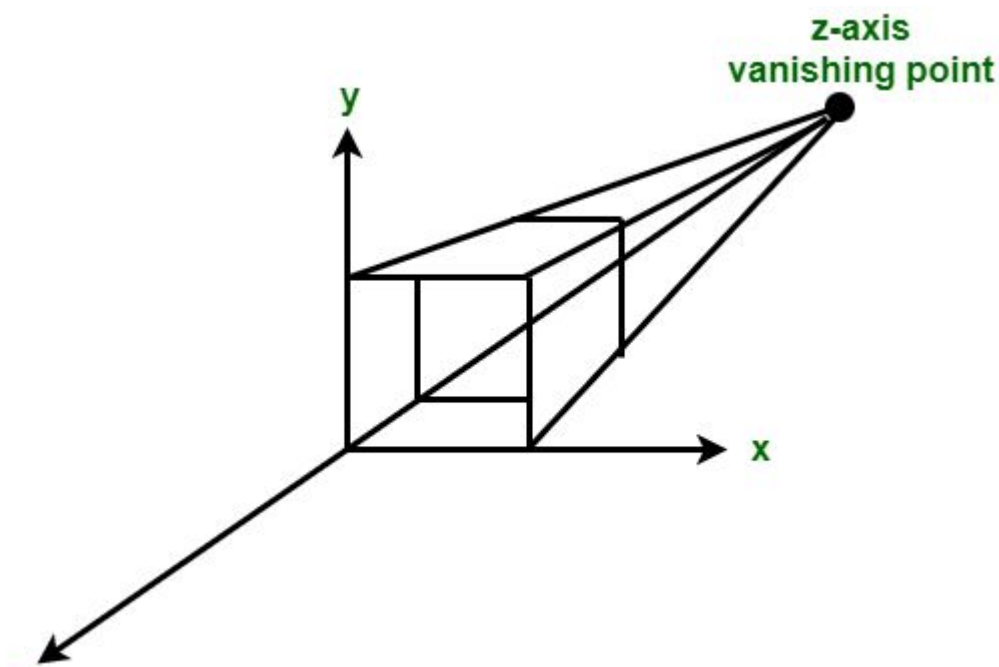
- **Center of Projection –** It is a point where lines or projection that are not parallel to projection plane appear to meet.
- **View Plane or Projection Plane –** The view plane is determined by :
  - View reference point $R_0(x_0, y_0, z_0)$
  - View plane normal.
- **Location of an Object –** It is specified by a point P that is located in world coordinates at (x, y, z) location. The objective of perspective projection is to determine the image point P' whose coordinates are (x', y', z')


Object
View
Plane
Converged
Projection lines
Centre of
Projection


Types of Perspective Projection
One-Point Perspective
Two-Point Perspective
Three-Point Persepective

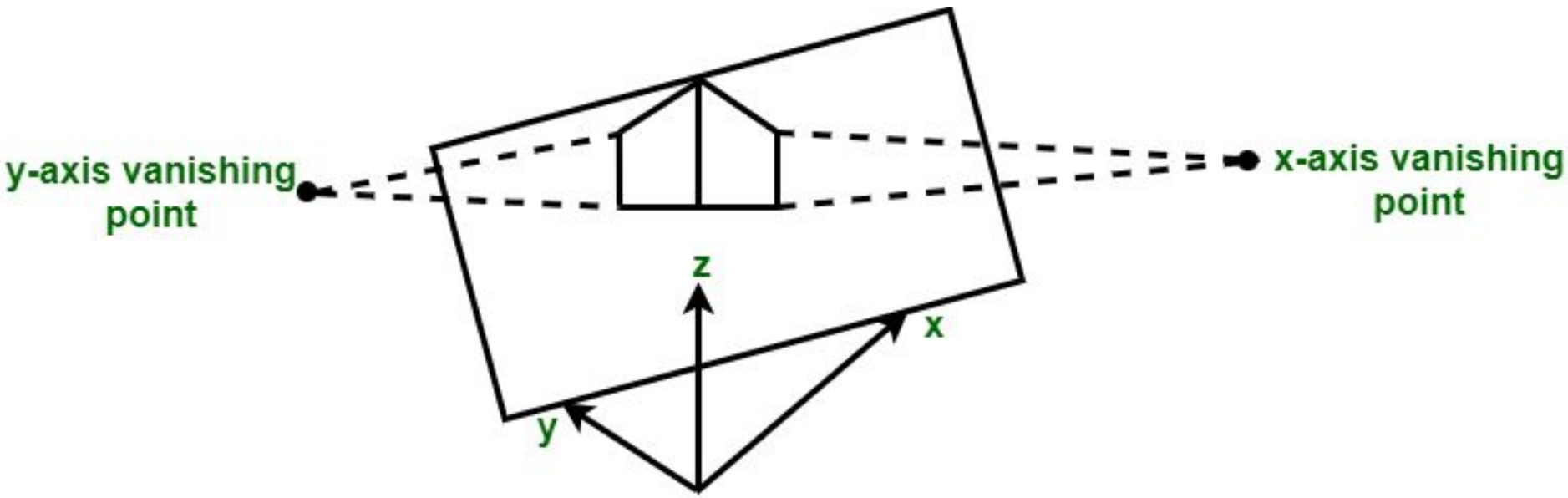
• 2.1) One Point Perspective Projection

- One point perspective projection occurs when any of principal axes intersects with projection plane or we can say when projection plane is perpendicular to principal axis.



- In the above figure, z axis intersects projection plane whereas x and y axis remain parallel to projection plane.

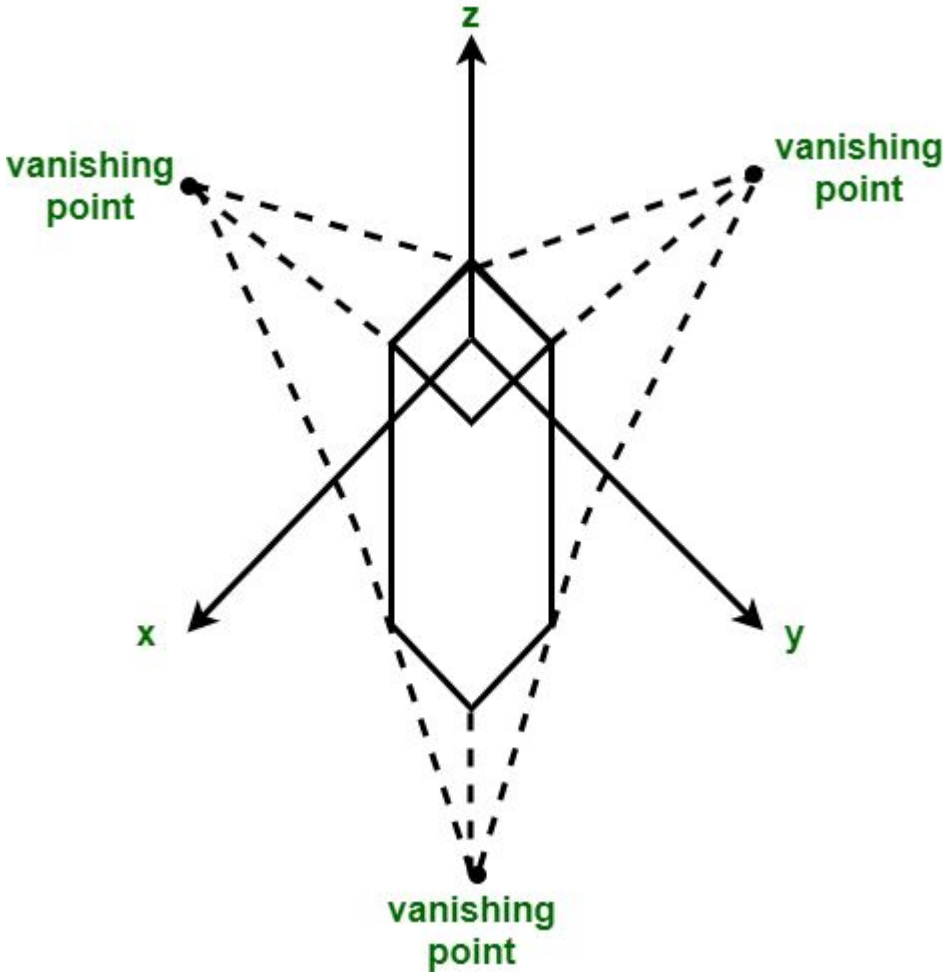
- **2.2)Two Point Perspective Projection**
- Two point perspective projection occurs when projection plane intersects two of principal axis.



- In the above figure, projection plane intersects x and y axis whereas z axis remains parallel to projection plane.

• 2.3)Three Point Perspective Projection

- Three point perspective projection occurs when all three axis intersects with projection plane.
- There is no any principal axis which is parallel to projection plane.



Application of Perspective Projection :

- The perspective projection technique is used by artists in preparing drawings of three-dimensional objects and scenes.

#Exemplar/Case Studies

1) Affine Transformations Vlab

(Vlab link: https://cse19-iiith.vlabs.ac.in/exp/affine-transformation/theory.html)

2) Image augmentation in Deep learning

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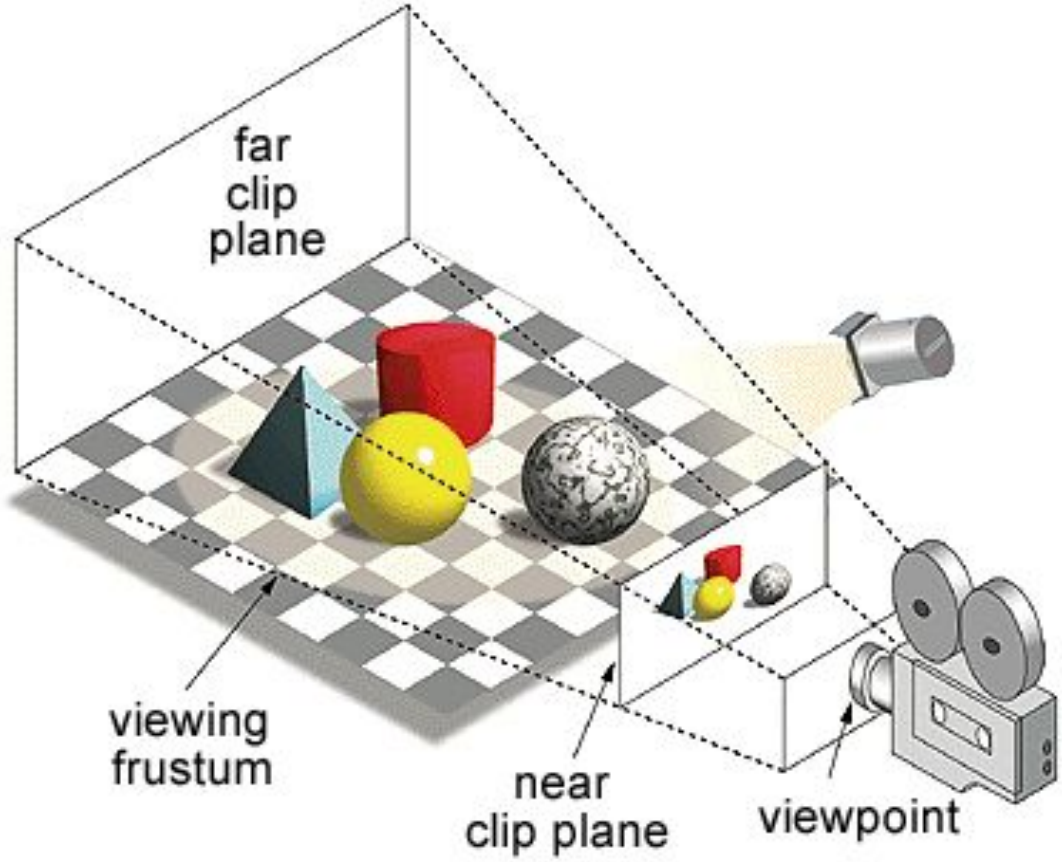
Thank You !!!